GCI World

Model Evaluation

松尾・岩澤研究室
MATSUO-IWASAWA LAB UTOKYO

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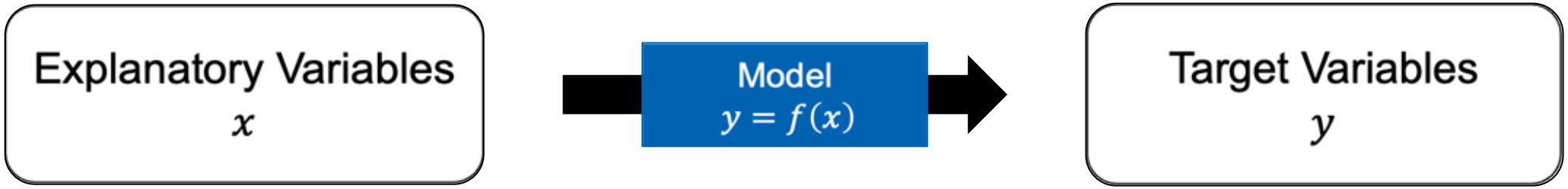
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Review of Supervised Learning

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Supervised Learning

- Learn the relationship f between the input x and output y



- e.g. image recognition, fraud detection, trend prediction, etc.

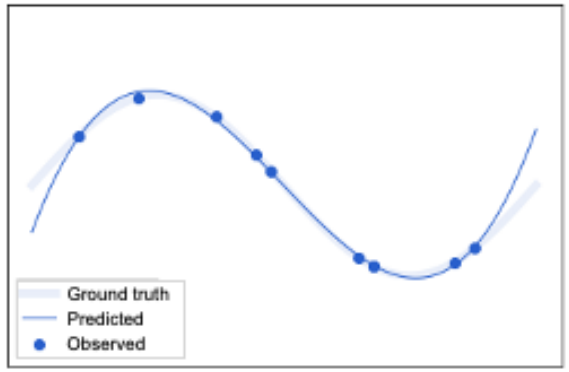
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Types of machine learning (1): Regression

Regression

Predicting continuous values

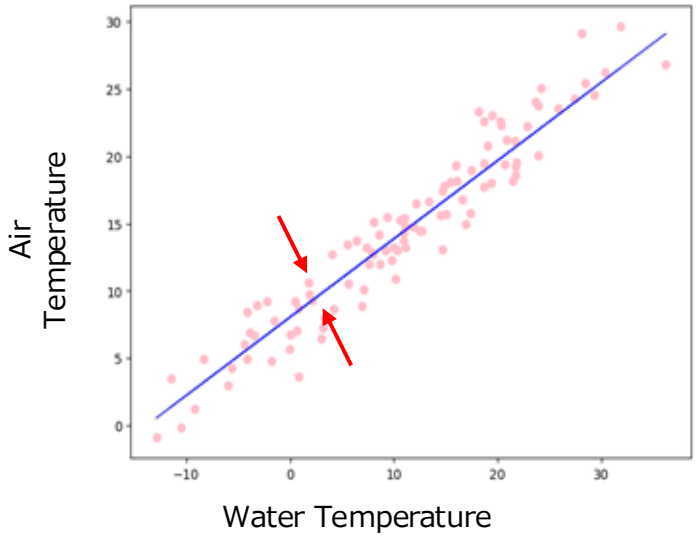
- e.g., house price prediction, temperature forecast



Algorithms

- Linear regression
- Ridge/Lasso regression
- Decision tree regression

e.g., Linear Regression



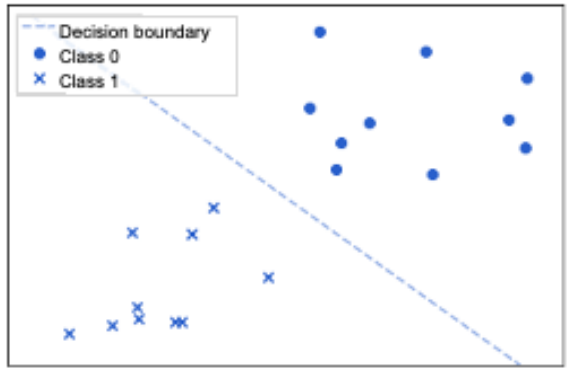
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Types of machine learning (2): Classification

Classification

Predicting discrete values (labels)

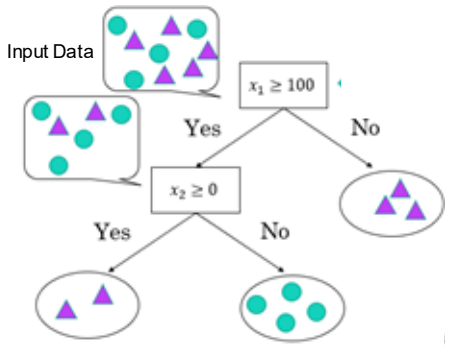
- e.g., predicting if email is sapam or not



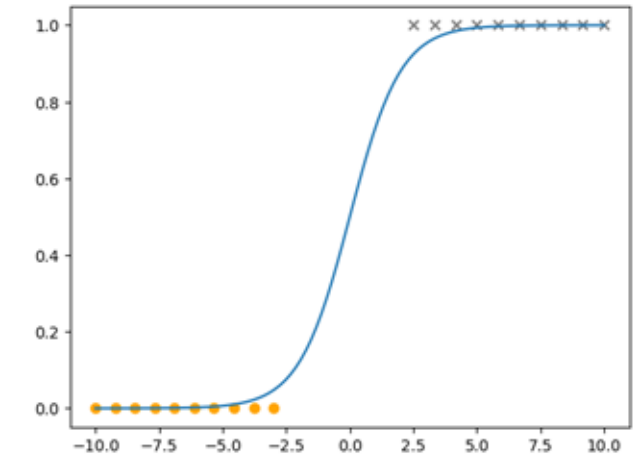
Algorithms

- Logistic regression
- Decision trees
- SVM

Decision Tree



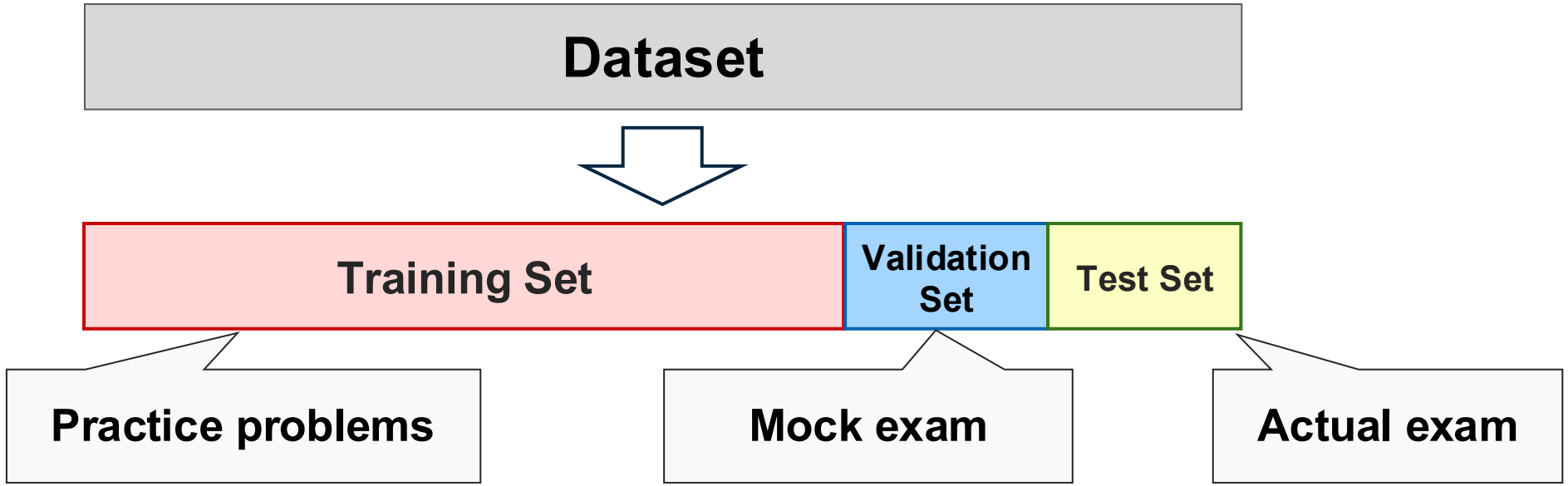
Logistic Regression



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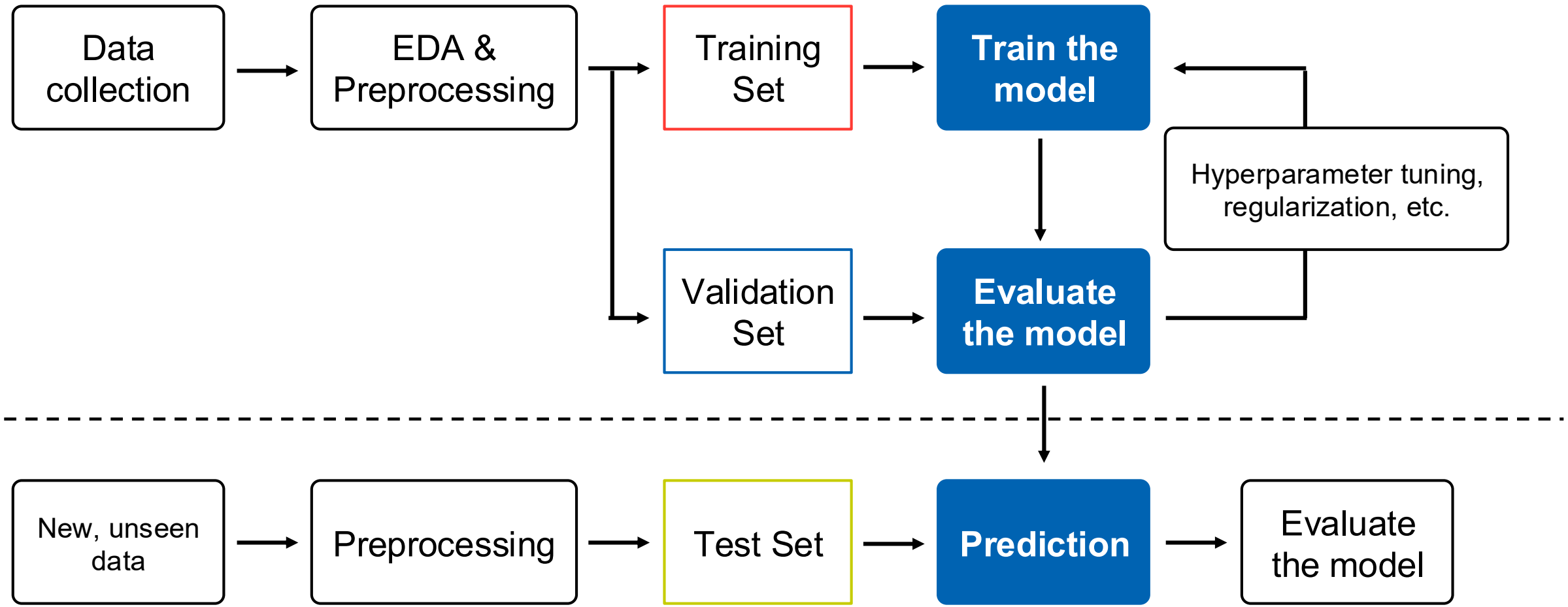
How to Evaluate the Model

- Split the data into
  - **Training set**: Use to train the model
  - **Validation set**: Use to evaluate the model
  - **Test set**: Use to evalute the best model



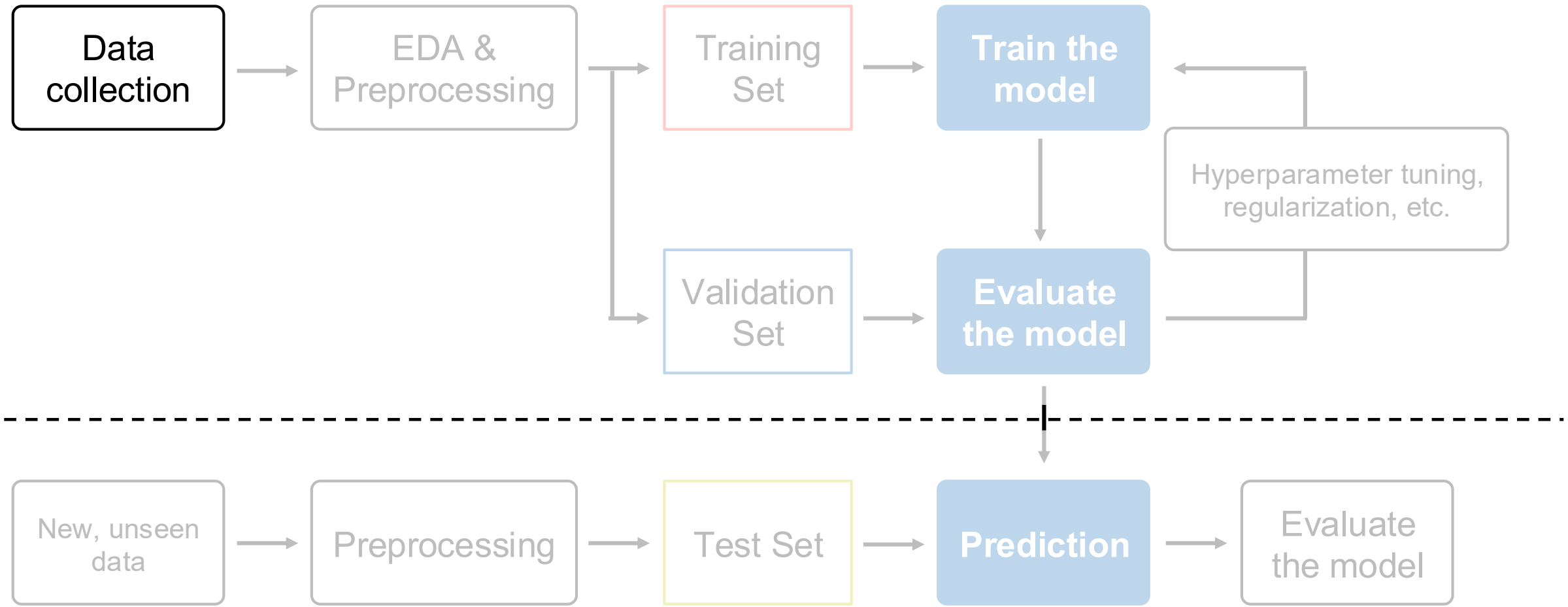
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Supervised Learning Workflow (Summary)



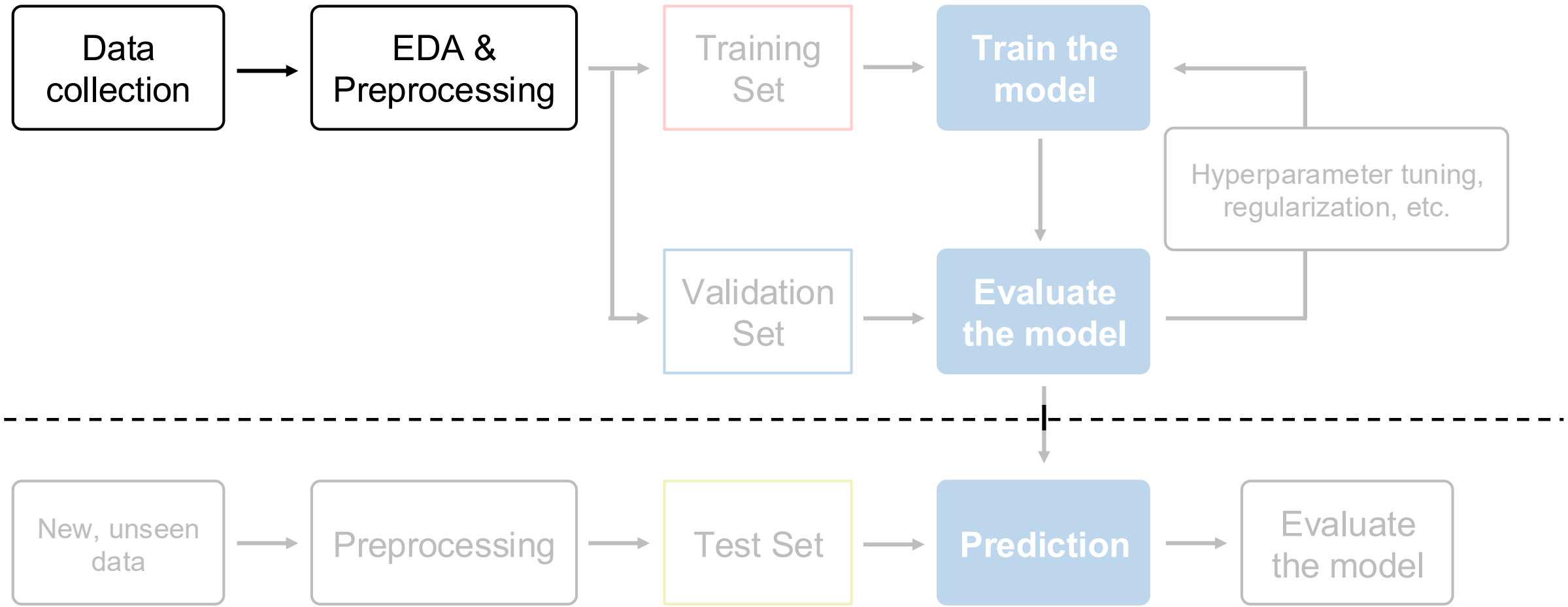
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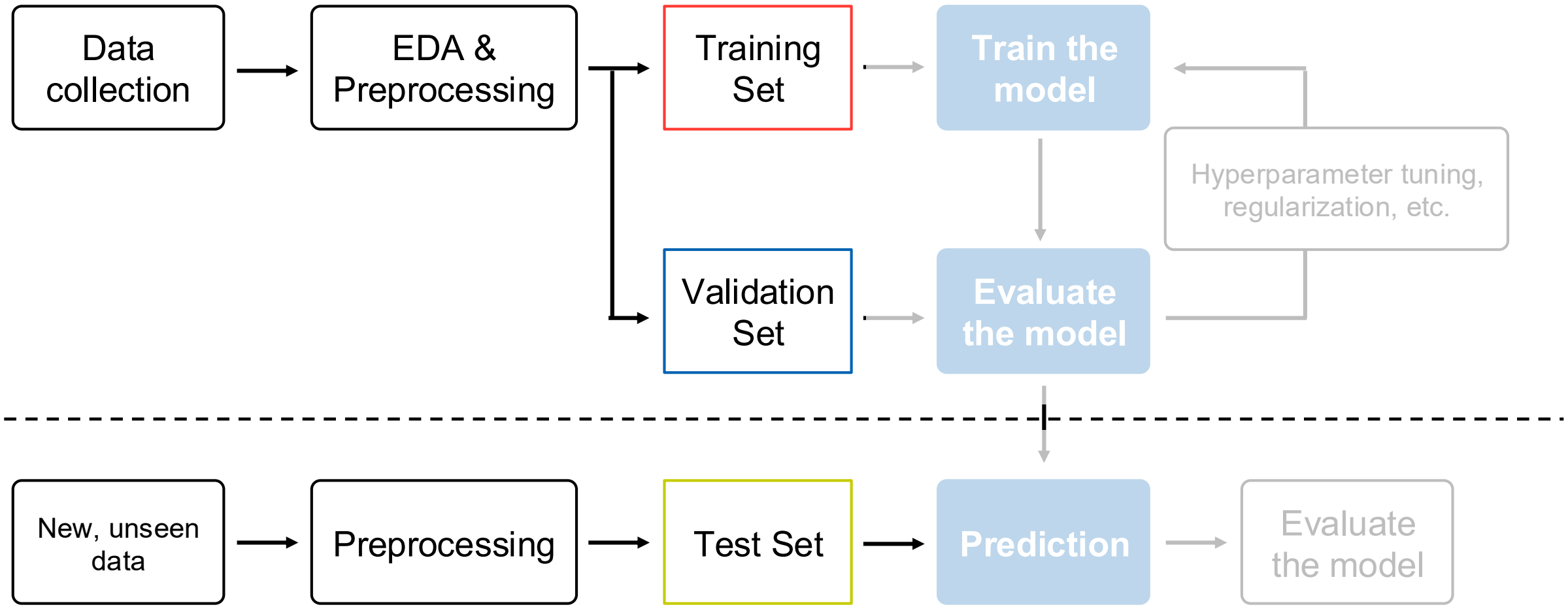
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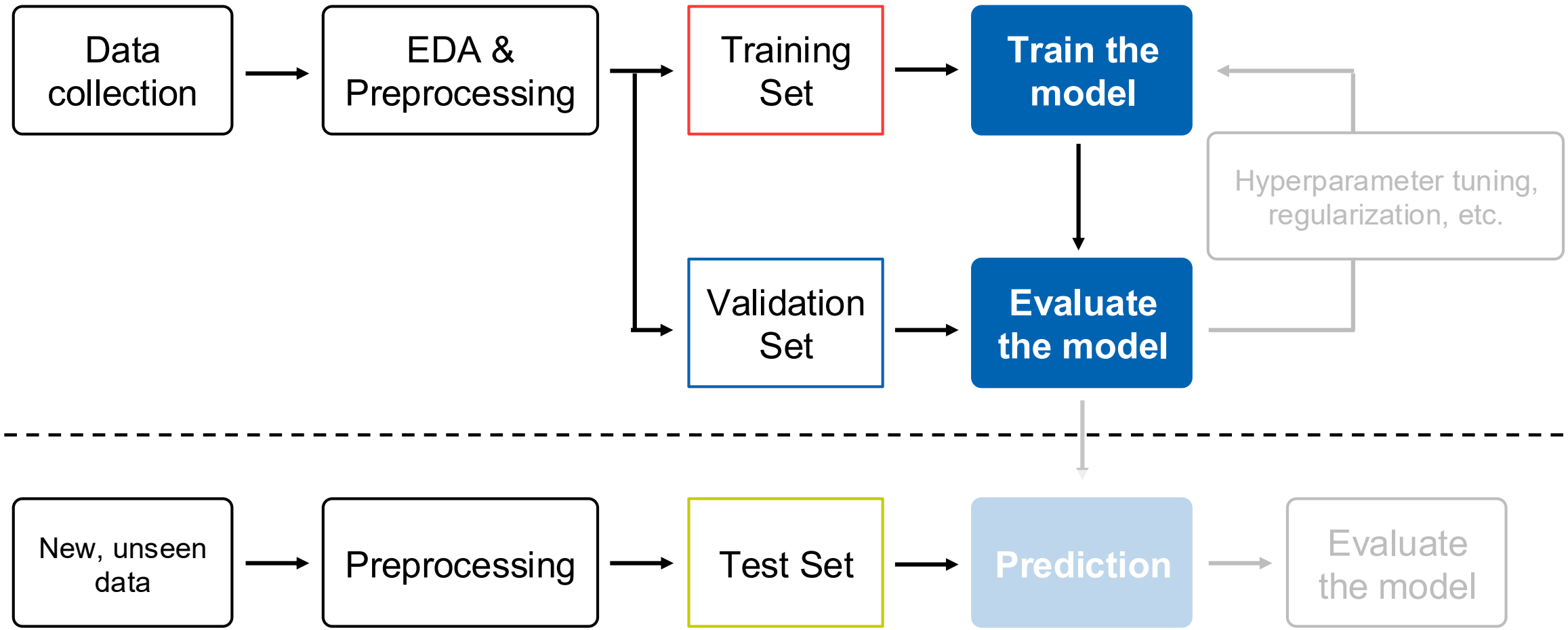
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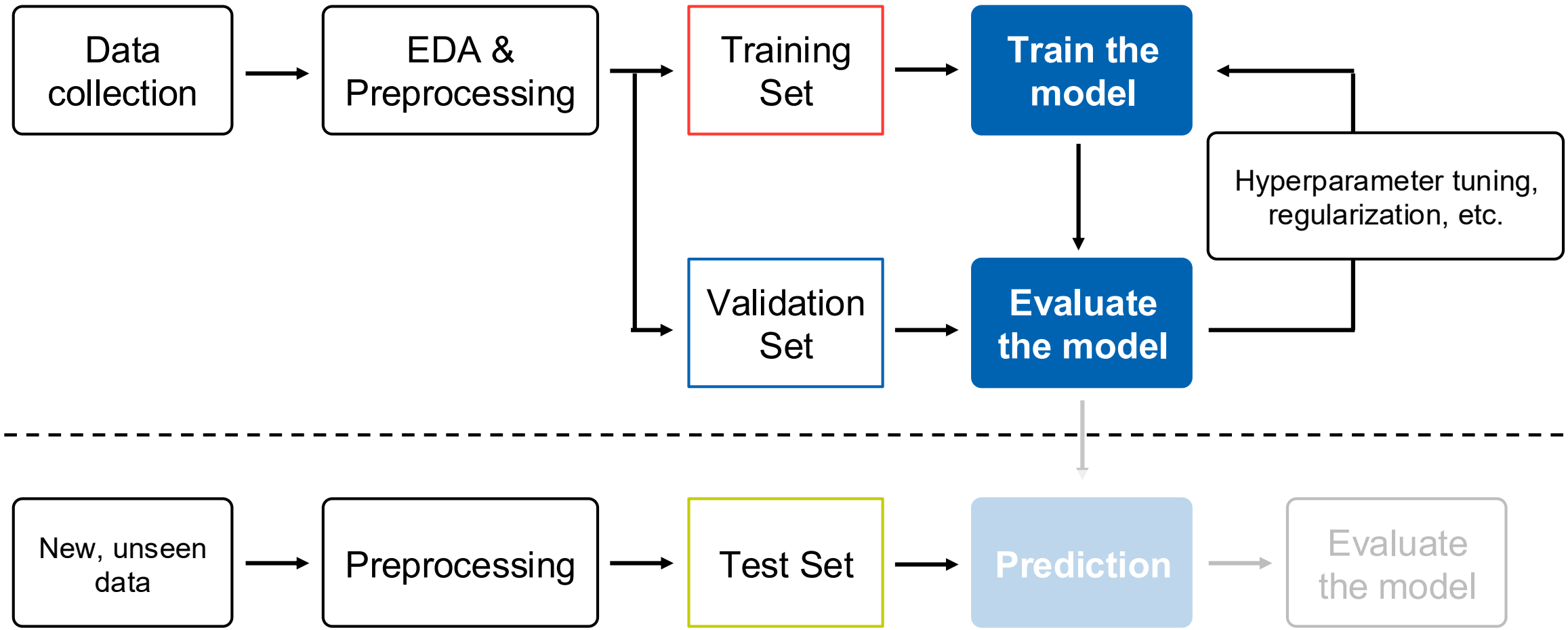
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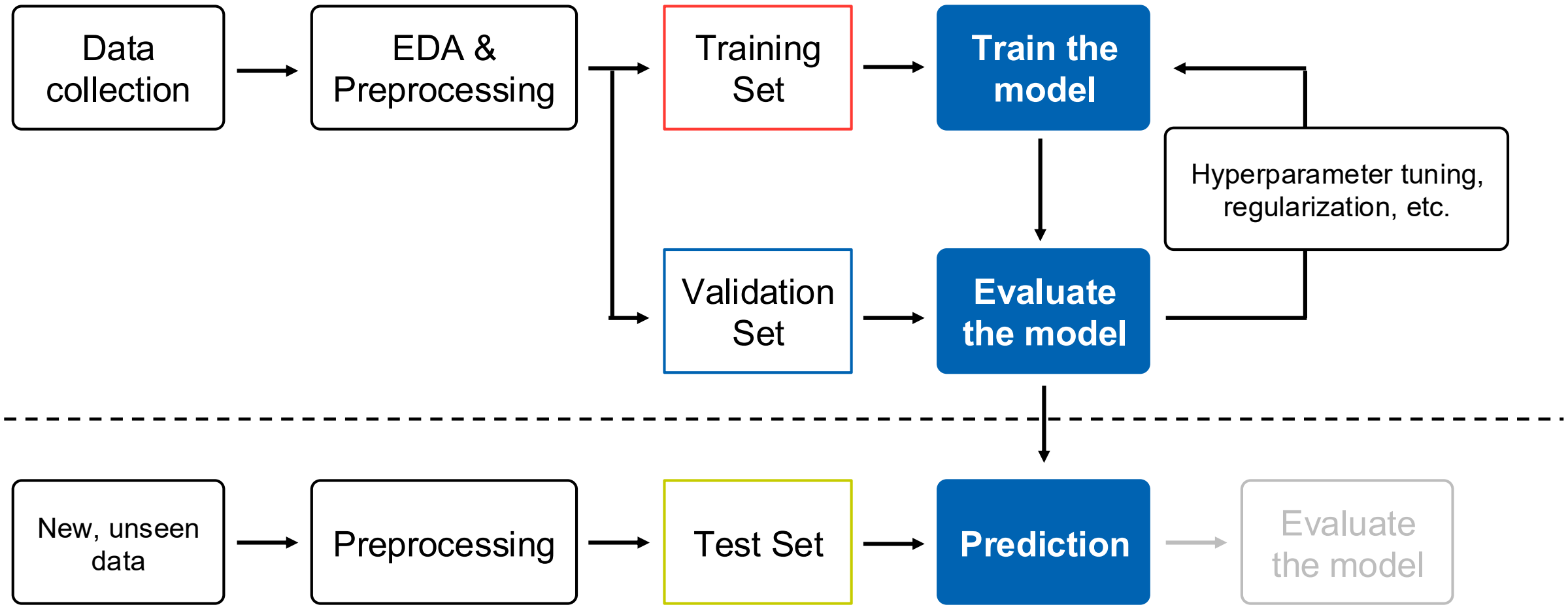
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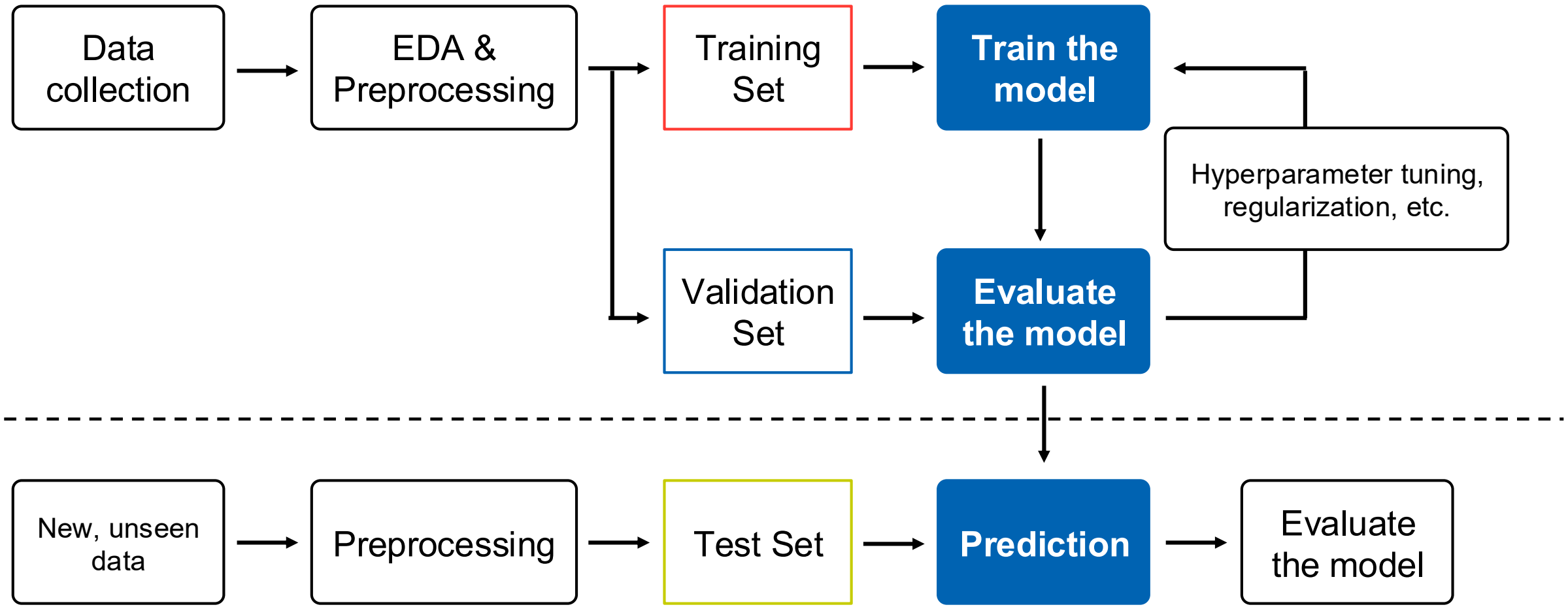
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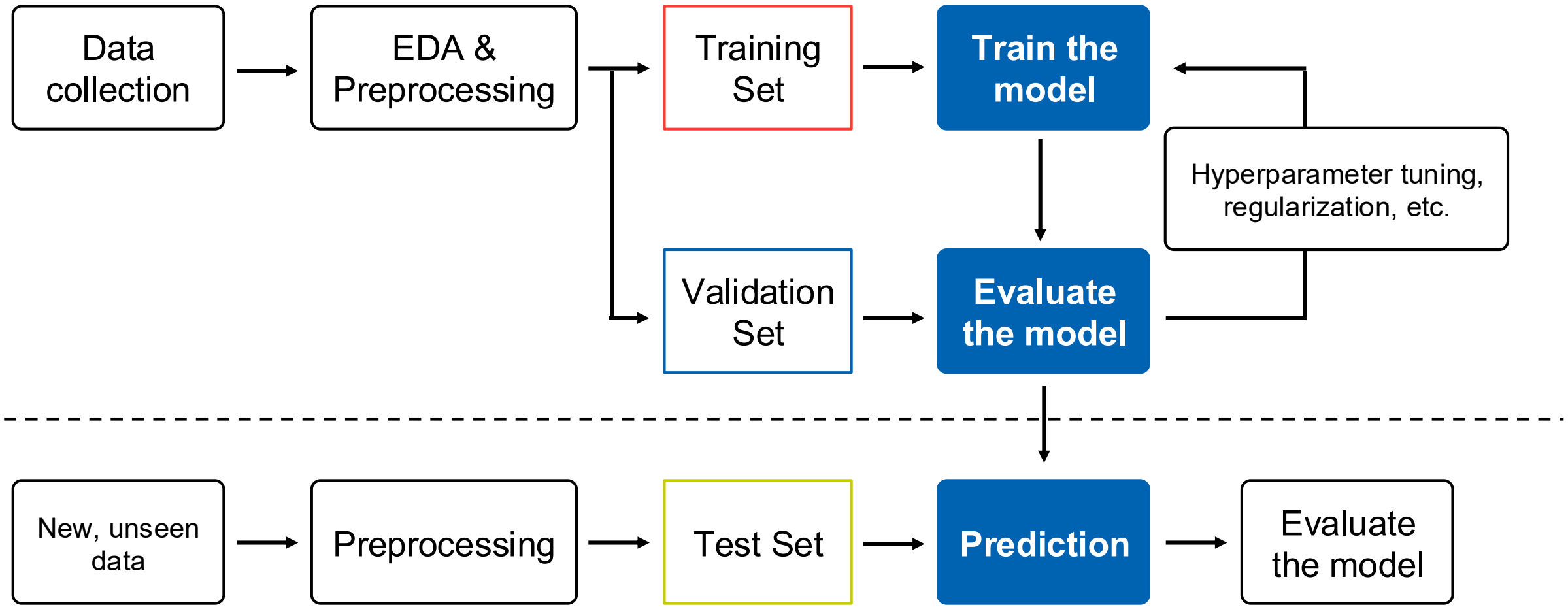


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Overview of Model Evaluation

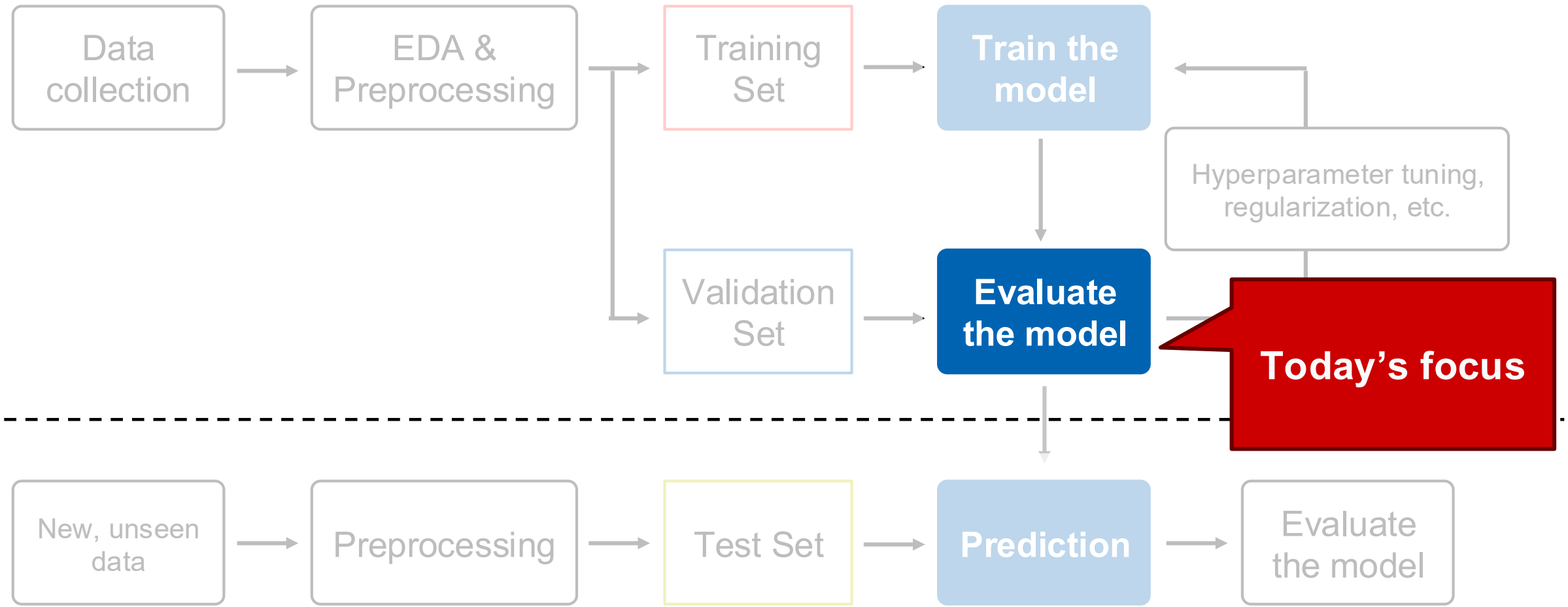
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Supervised Learning Workflow (Summary)



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What is a good model?

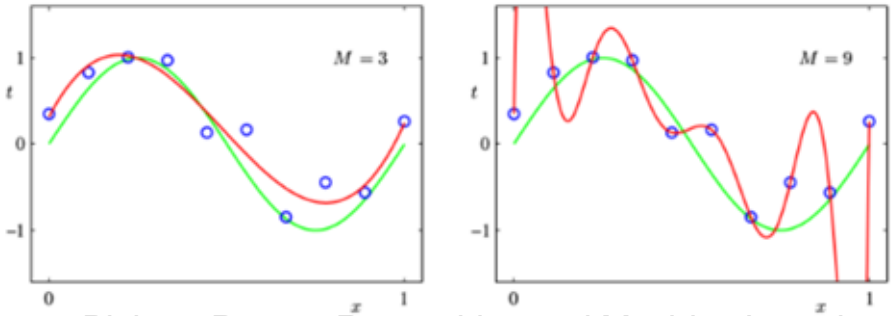
- Why do we need the evaluation step?

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What is a good model?

- Why do we need the evaluation step?

- To see the model generalizes well.
  - **Generalization**: good performance on unseen data
  - We must avoid **overfitting**: rote learning of the training data

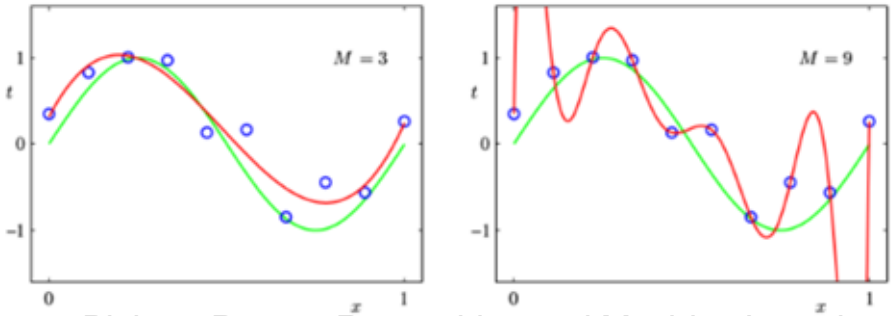


Bishop, Pattern Recognition and Machine Learning

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Bishop, Pattern Recognition and Machine Learning

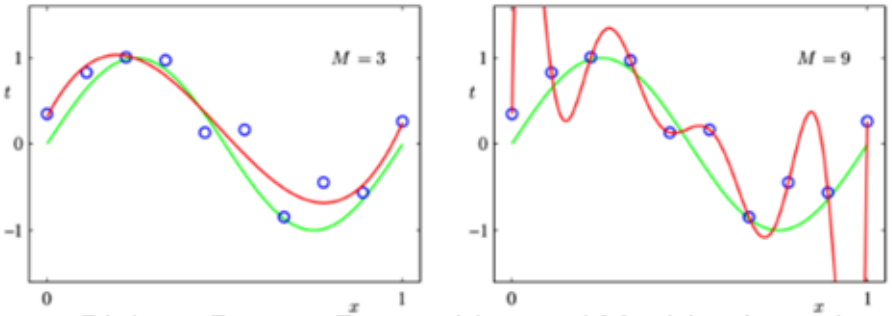
- To see the model generalizes well.
  - **Generalization**: good performance on unseen data
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- To choose a good hyperparameter.
  - Hyperparameter(s): the values needed to be set before training
  - e.g., normalization coefficient (λ), tree depth, etc.

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What is a good model?

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Bishop, Pattern Recognition and Machine Learning

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What metrics to use in evaluating your model?

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Evaluation for Classification Models

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Case Study: disease diagnosis system of high accuracy

- “We’ve created a disease diagnosis test of 99% accuracy!”

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Case Study: disease diagnosis system of high accuracy

- “We’ve created a disease diagnosis test of 99% accuracy!”



Can we really trust this test?

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Case Study: disease diagnosis system of high accuracy

- “We’ve created a disease diagnosis test of 99% accuracy!”

Truth		
	Negative/0	9990
	Positive/1	10

Data seems highly deviated (negative >> positive).

In such situations, can we evaluate the test just by accuracy?

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VERY IMPORTANT Confusion Matrix

- Confusion matrix: a table of how many true positive/negative population were predicted positive/negative

		Prediction	
		Negative/0	Positive/1
Ground Truth	Negative/0	TN True Negative	FP False Positive
	Positive/1	FN False Negative	TP True Positive

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Accuracy

- % of predicted and correct values match, **regardless of the deviation of the correct value**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

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		Prediction	
		Negative/0	Positive/1
Truth	Negative/0	True Negative 9900	False Positive 90
	Positive/1	False Negative 8	True Positive 2

(2 + 9900) / (2 + 9900 + 90 + 8) = 99.0% Accuracy

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- If true negative ≫ true positive, the accuracy can be high when the model almost always predicts negative → **apparent high accuracy**

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Recall (sensitivity)

- % of cases correctly predicted as positive (TP) out of truly positive cases (TP+FN)

$$Recall = \frac{TP}{TP + FN}$$

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2 / (2 + 8) = **20.0% Recall**

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Recall (sensitivity)

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2 / (2 + 8) = **20.0% Recall**

- An indicator to see whether unmissable events can be detected

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Precision (Positive predictive value)

- % of cases that were actually positive (TP) out of those predicted to be positive (TP + FP)

$$Precision = \frac{TP}{TP + FP}$$

		Prediction	
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2 / (2 + 90) = **2.2% Precision**

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2 / (2 + 90) = **2.2% Precision**

- An indicator that measures the **cost-effectiveness of actions taken in accordance with predicted results**, such as when implementing measures in response to a positive example would be high cost.

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F-Value（F1）

- The “harmonic mean” of fit and repeatability

$$F_1 = \frac{2}{\frac{1}{recall} + \frac{1}{precision}} = \frac{2TP}{2TP + FP + FN}$$

		Prediction	
		Negative/0	Positive/1
Truth	Negative/0	True Negative 9900	False Positive 90
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2*2 / (2*2 + 90 + 8) = **3.92% F1**

- Indicator used to compare and evaluate machine learning models in terms of both goodness-of-fit and reproducibility

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How to implement

- Accuracy, precision, recall, and F1

```
from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
)

y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
```

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Your model’s prediction changes with threshold

- These metrics were based on model’s positive/negative prediction results.

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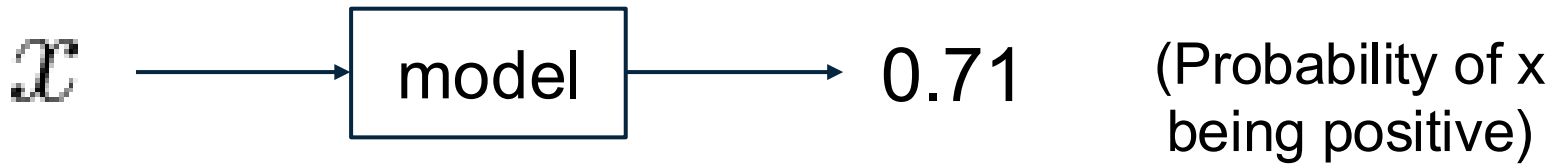
- However, what a machine learning model tells you is not just pos/neg
  – it tells you the **probability** of being positive.

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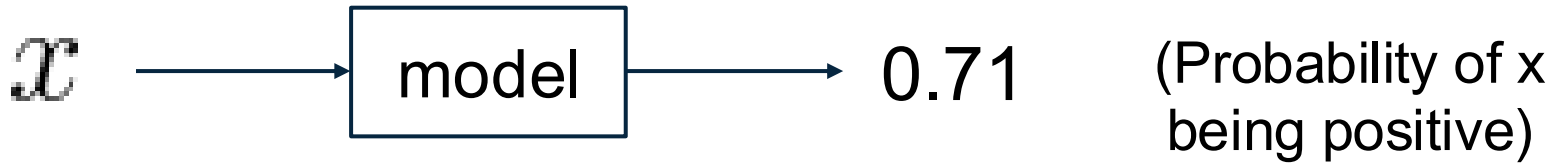


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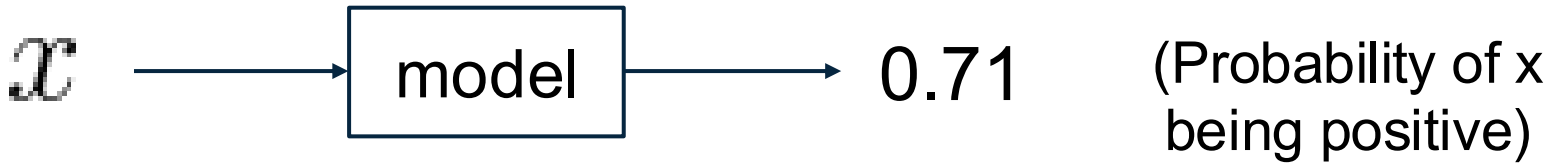
- And the model's pos/neg prediction changes depending on the **threshold** you set.

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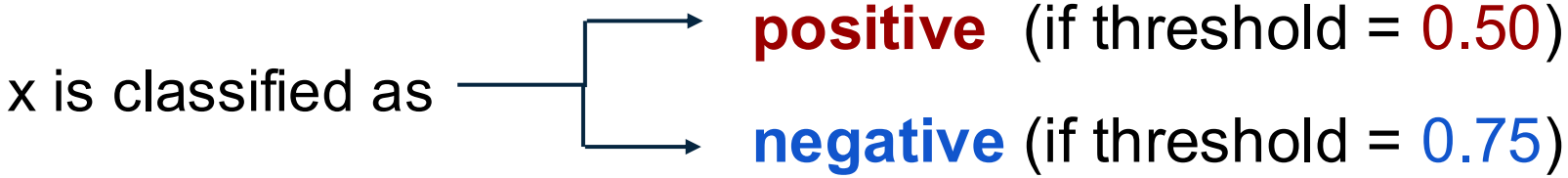
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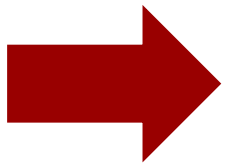
Your model's prediction changes with threshold

- Any metric that is **unaffected by the threshold**?
- Differently put, does your model nicely **rank true positives higher than true negatives**?

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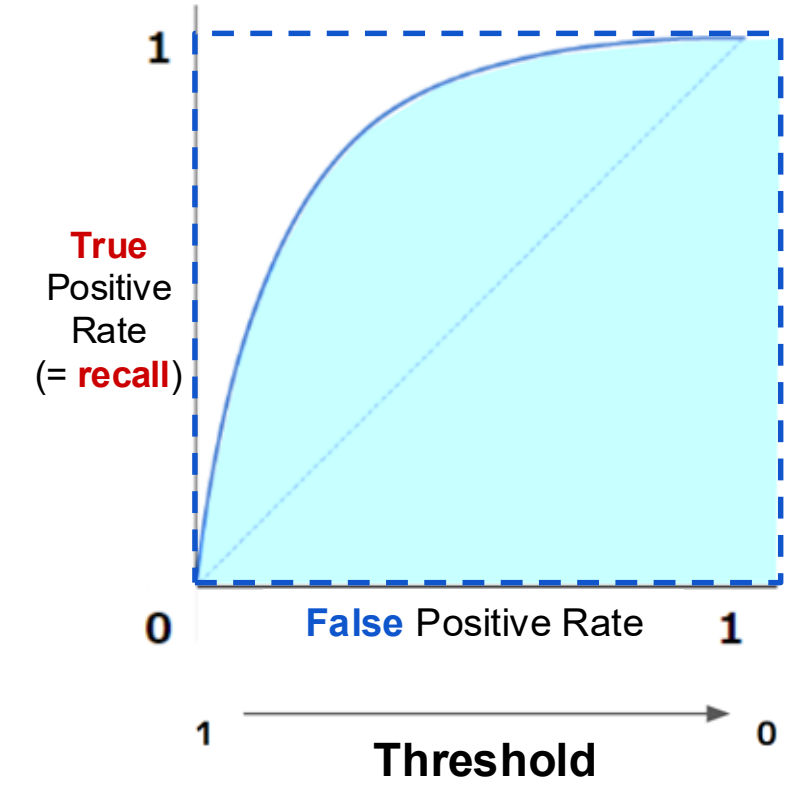


ROC (Receiver Operating Characteristic) curve
ROC-AUC (area under the curve)

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Evaluation of Classification Model: ROC and ROC-AUC

- ROC (Receiver Operating Characteristic) curve
  - A curve plotting the **true positive rates** (= **recall**) and **false positive rates** by predictive probability threshold.

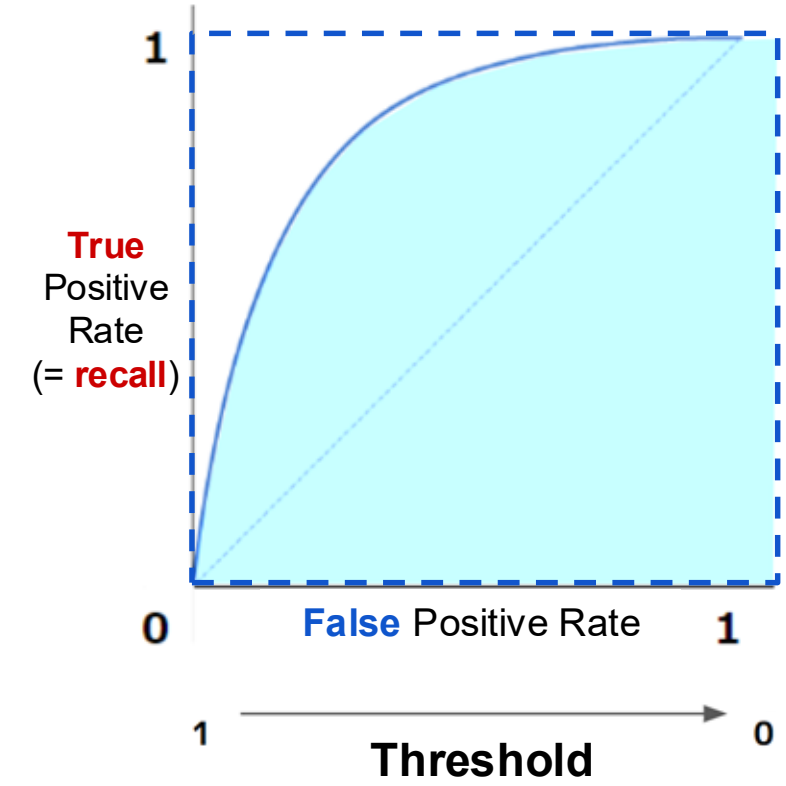


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Evaluation of Classification Model: ROC and ROC-AUC

- ROC (Receiver Operating Characteristic) curve
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- ROC-AUC (Area Under the Curve)
  - The area under the ROC curve (marked as light blue).

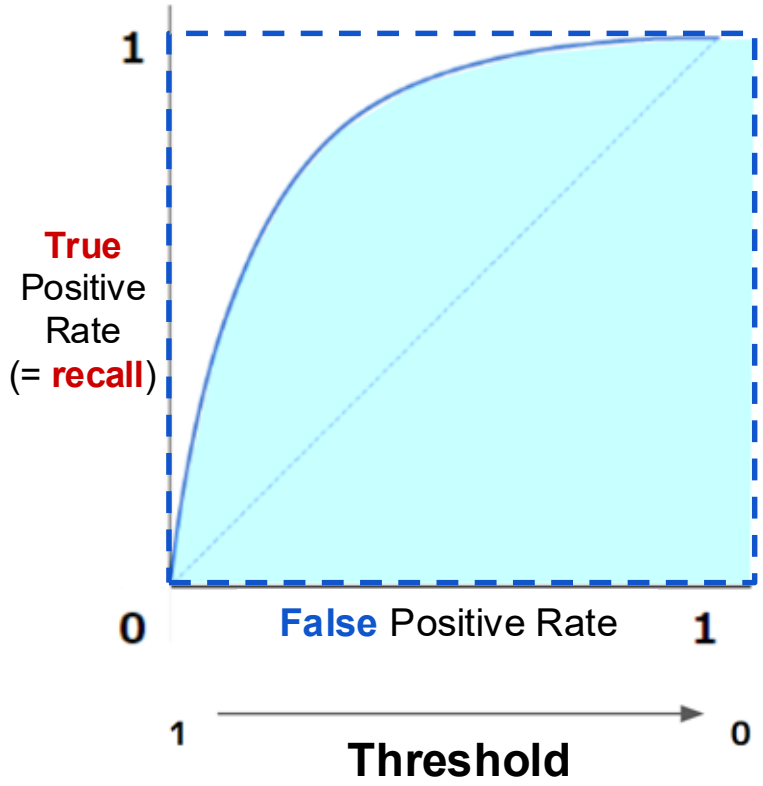


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Evaluation of Classification Model: ROC and ROC-AUC

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  - A curve plotting the **true positive rates** (= **recall**) and **false positive rates** by predictive probability threshold.

- ROC-AUC (Area Under the Curve)
  - The area under the ROC curve (marked as light blue).
  - $0 \leqq$ ROC-AUC $\leqq 1$.
  - A large ROC-AUC represents that the model is a good cl



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ROC curve

- How to draw an ROC curve?

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ROC curve

- How to draw an ROC curve?

1. Put all samples into your model and compute the probability of being positive (= score).

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ROC curve

- How to draw an ROC curve?

1. Put all samples into your model and compute the probability of being positive (= score).

ID	Truth	Probability
1	1	0.82
2	1	0.58
3	0	0.63
4	0	0.21
5	1	0.71
6	0	0.42
:	:	:

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ROC curve

- How to draw an ROC curve?

1. Put all samples into your model and compute the probability of being positive (= score).

2. Draw a histogram of scores over all positive & negative samples.

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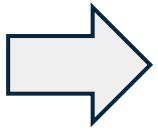
ROC curve

- How to draw an ROC curve?

1. Put all samples into your model and compute the probability of being positive (= score).

2. Draw a histogram of scores over all positive & negative samples.

ID	Truth	Probability
1	1	0.82
2	1	0.58
3	0	0.63
4	0	0.21
5	1	0.71
6	0	0.42
:	:	:



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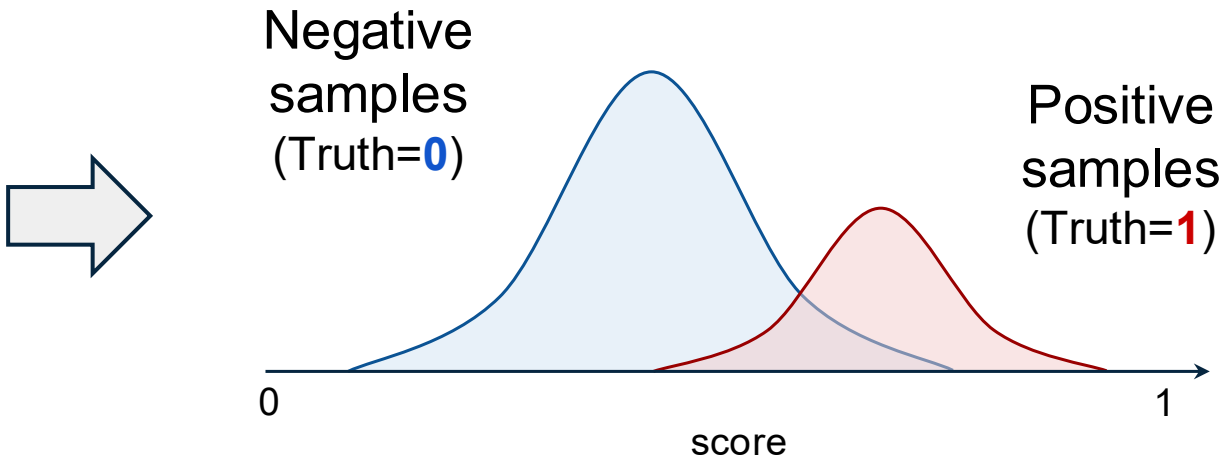
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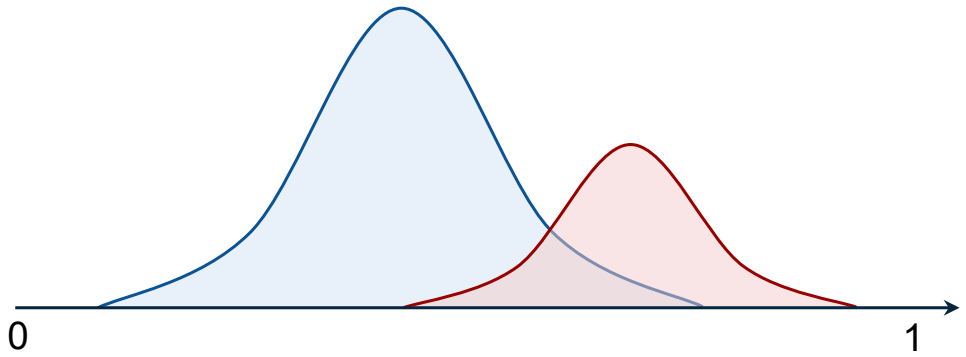
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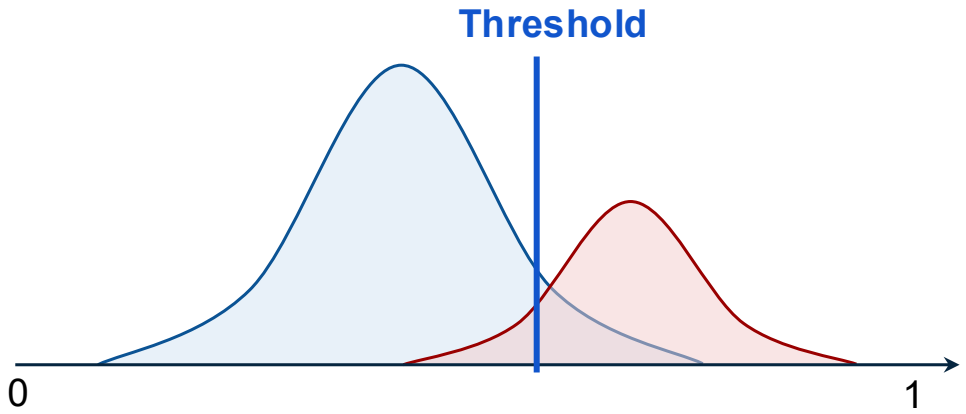
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ROC curve



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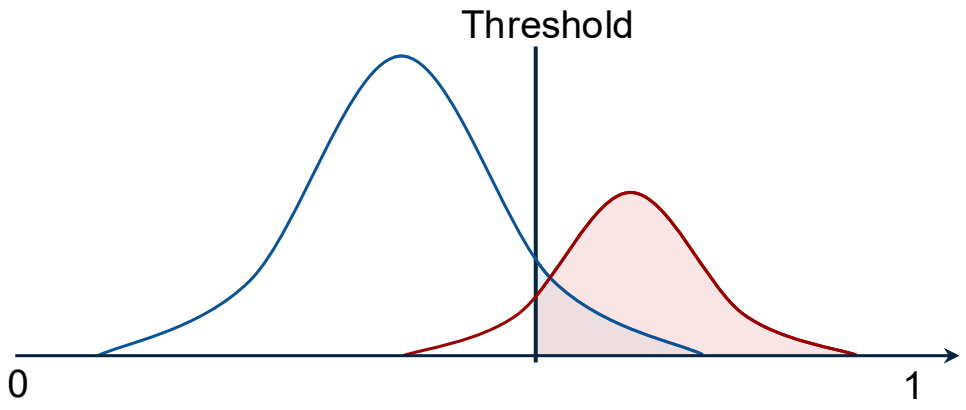
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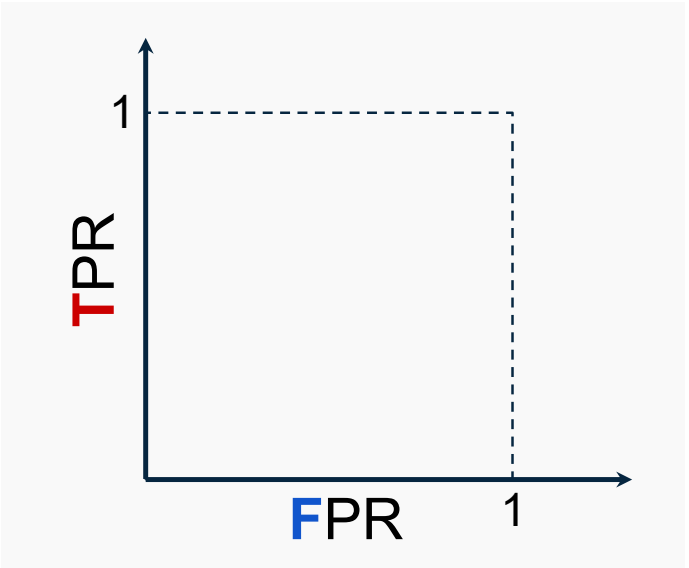


3. Set a **threshold** and compute true positive rate (TPR) and false positive rate (FPR). Plot the TPR and FPR on an xy plane.

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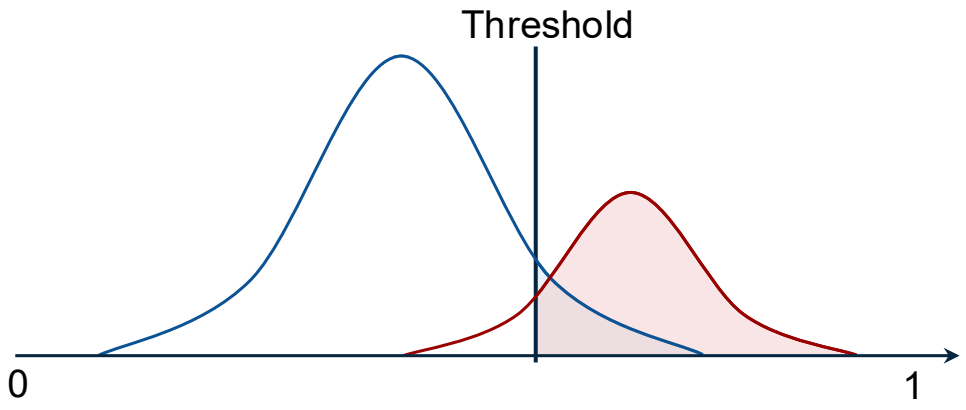


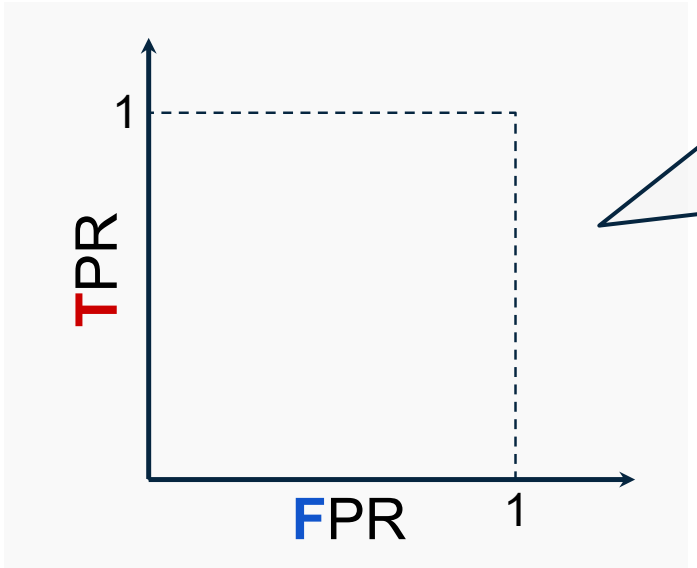


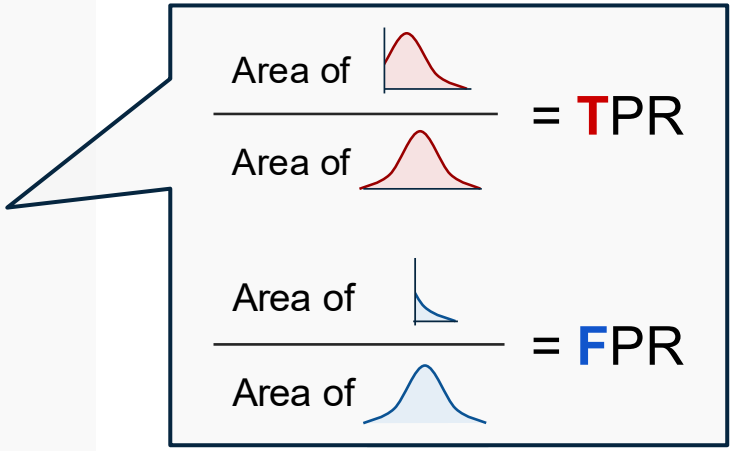
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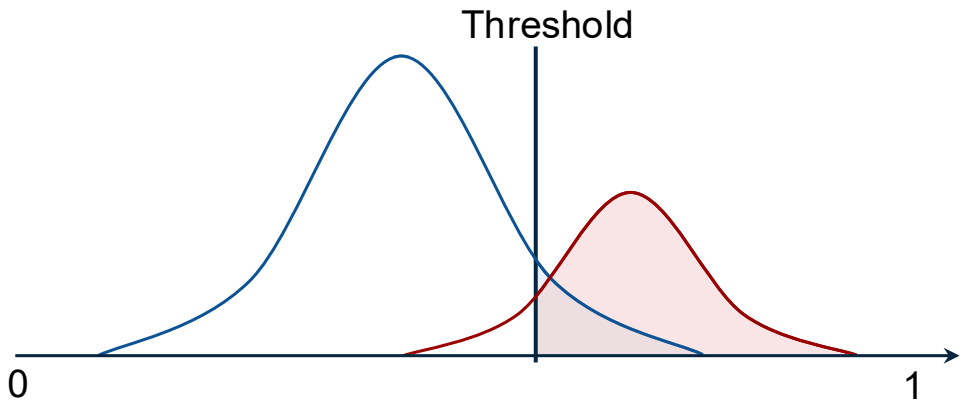


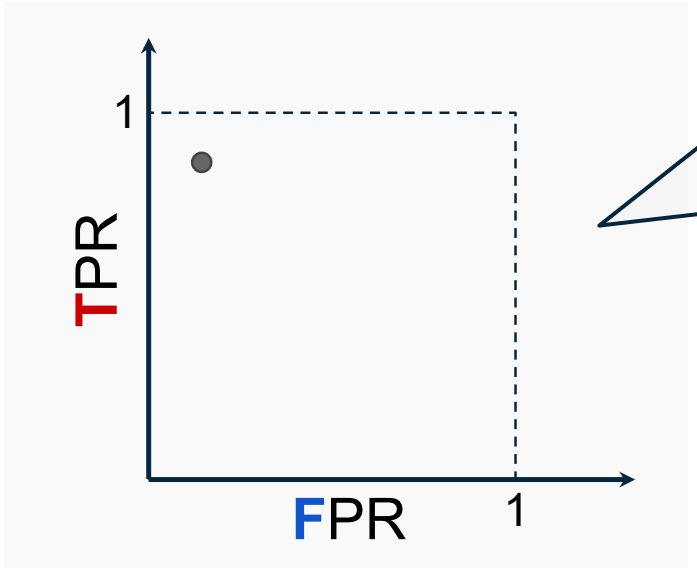


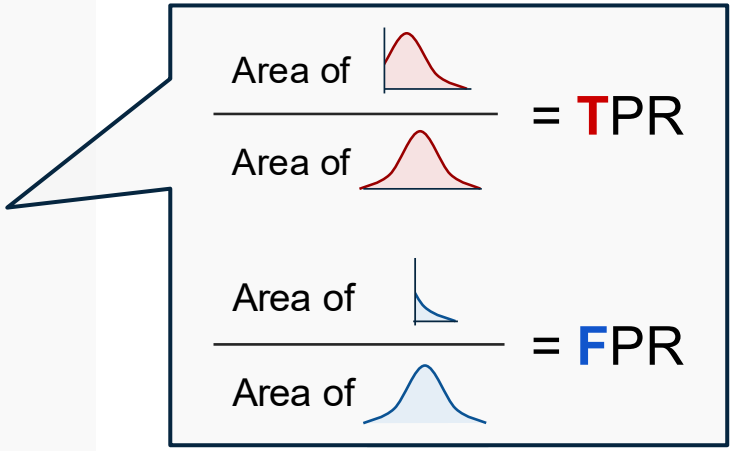
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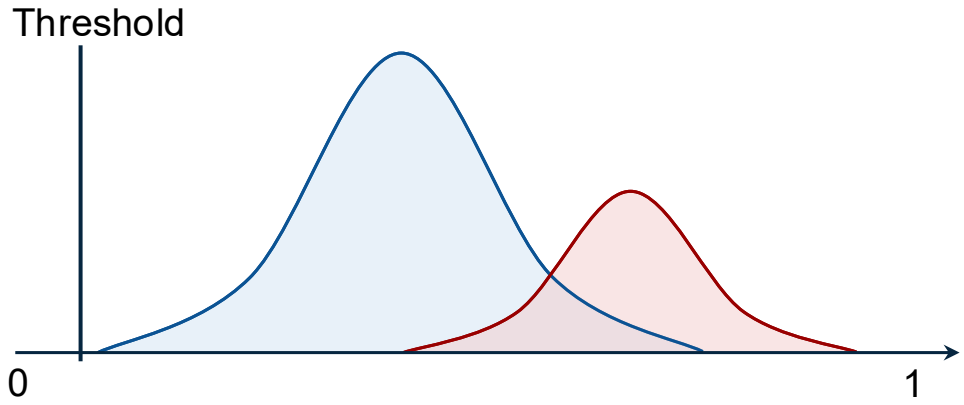


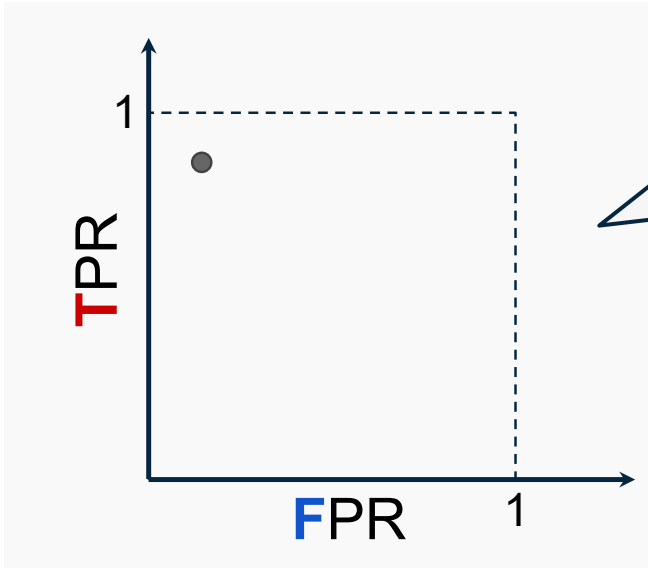


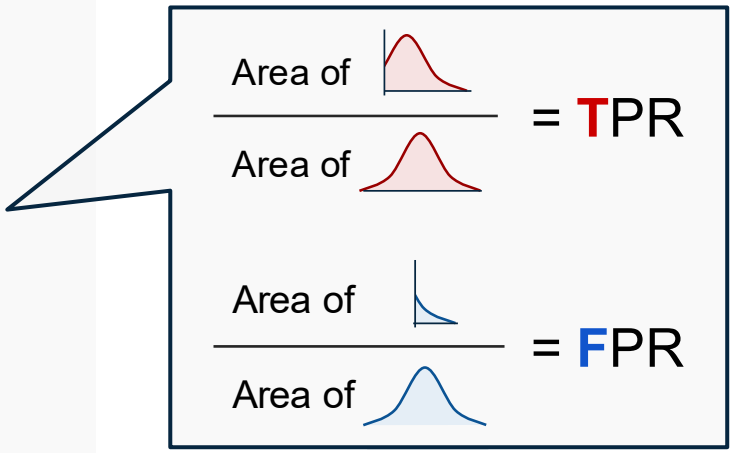
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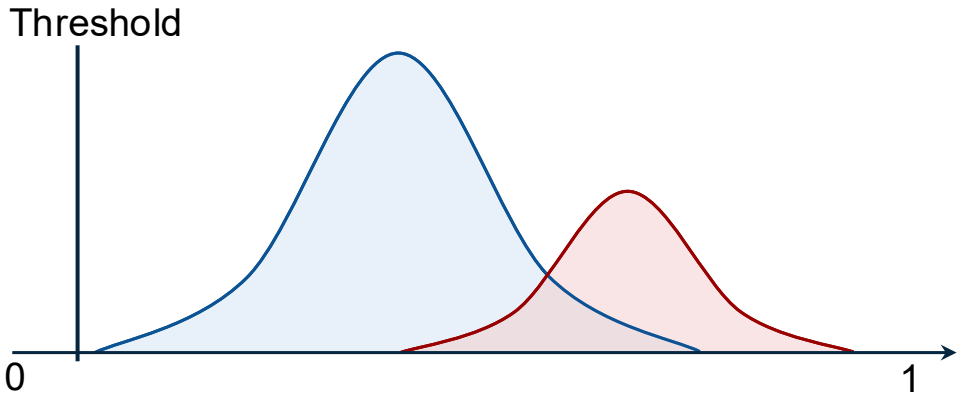


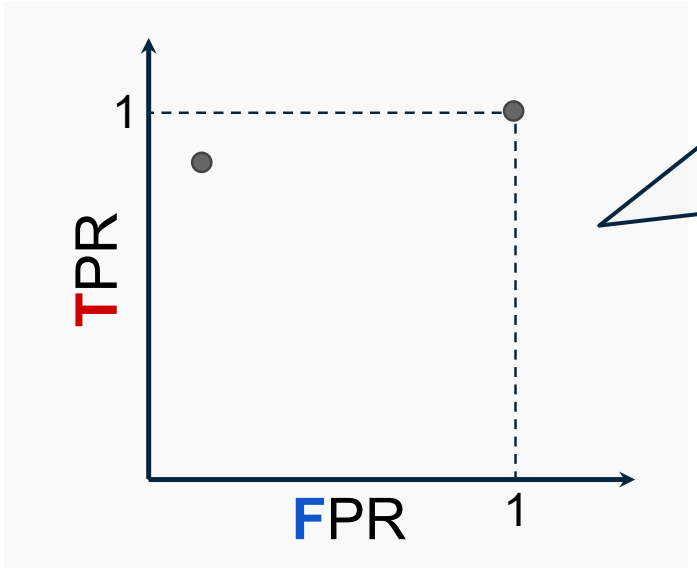


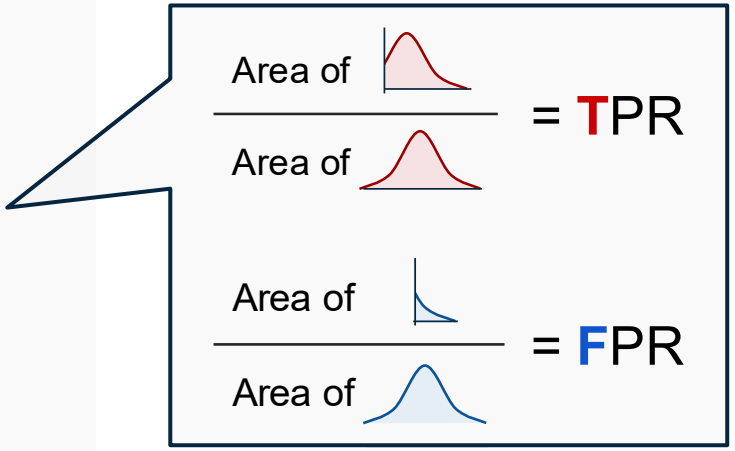
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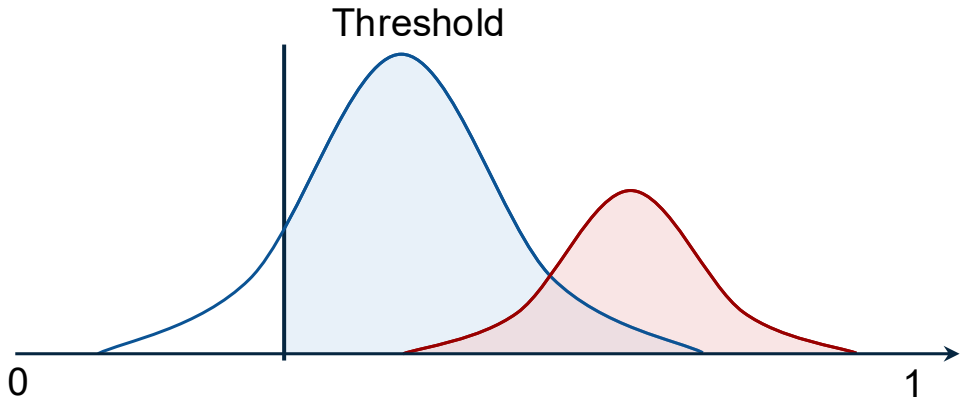


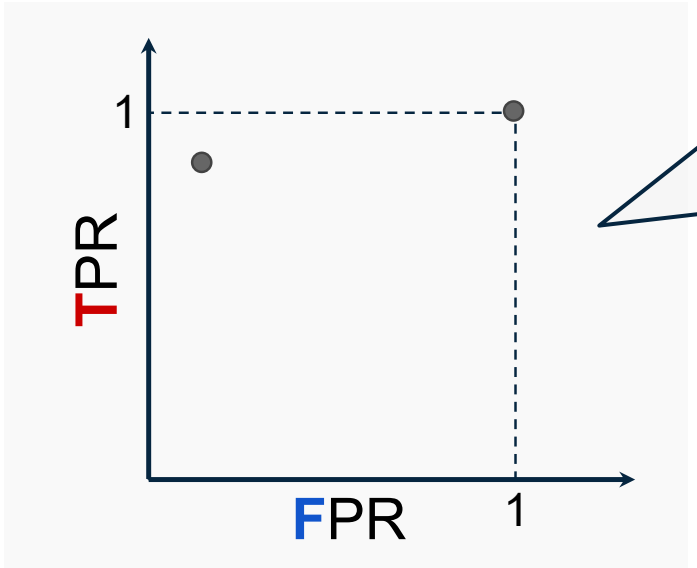


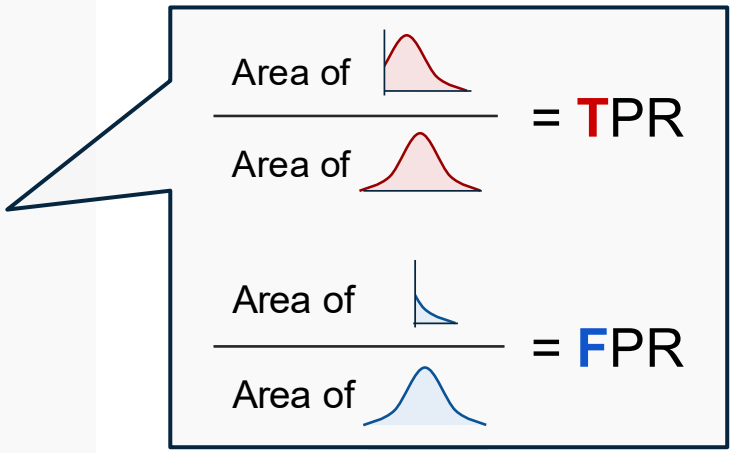
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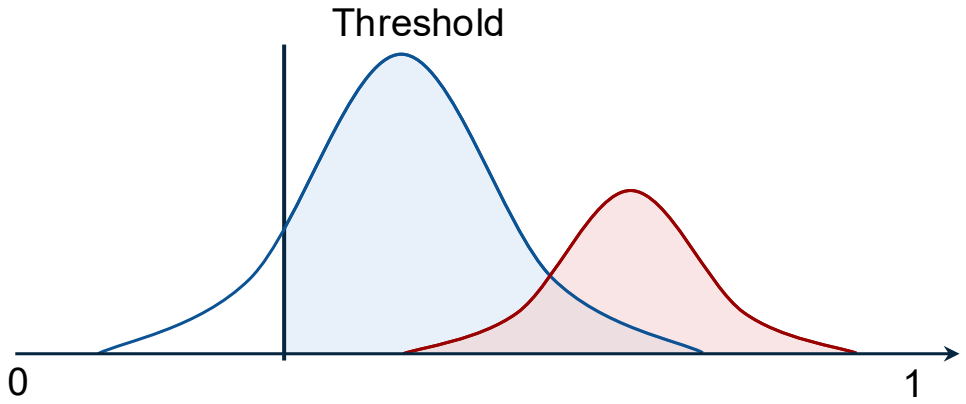


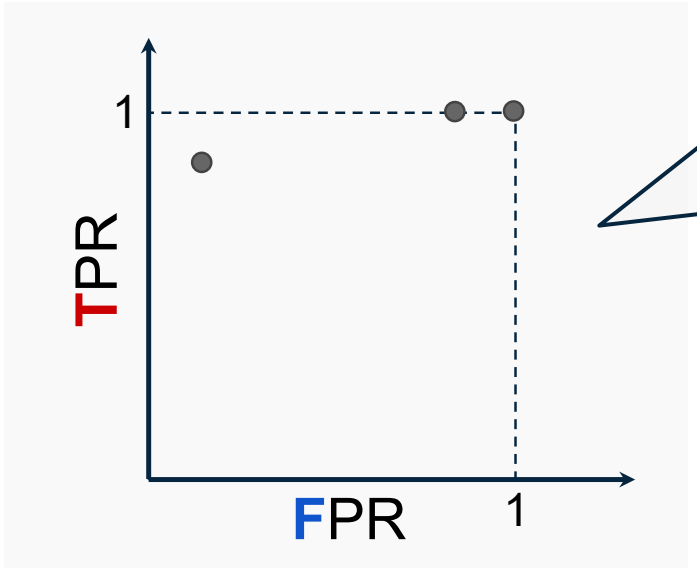


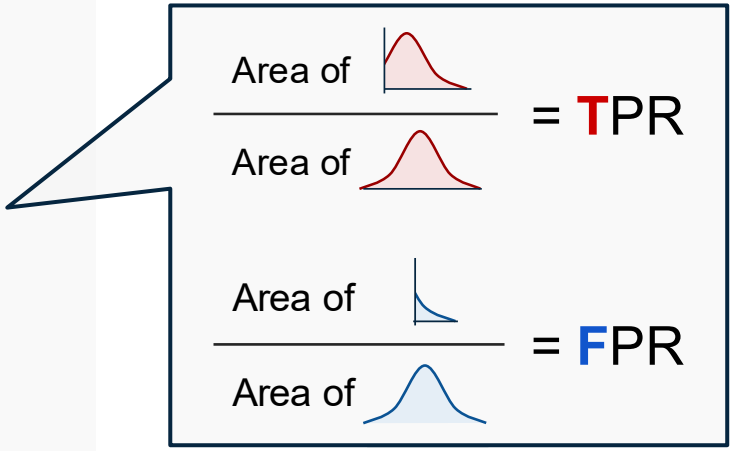
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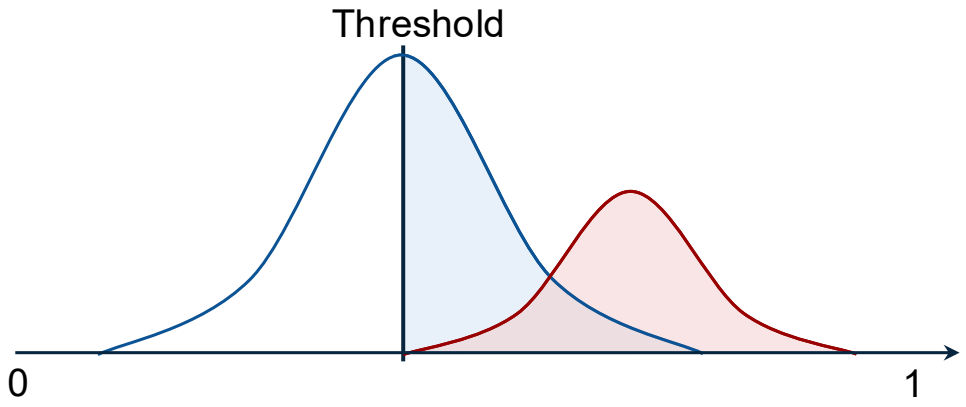


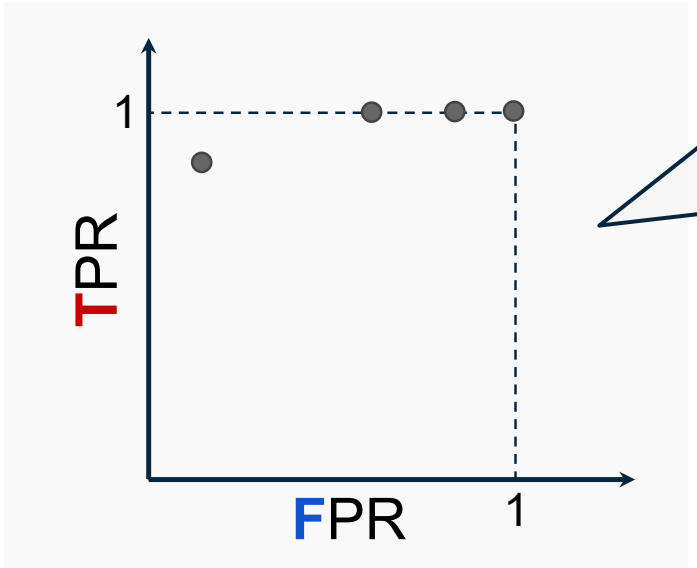


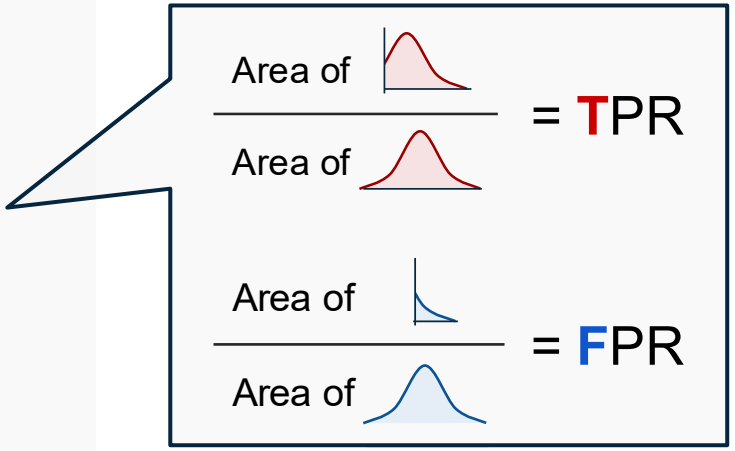
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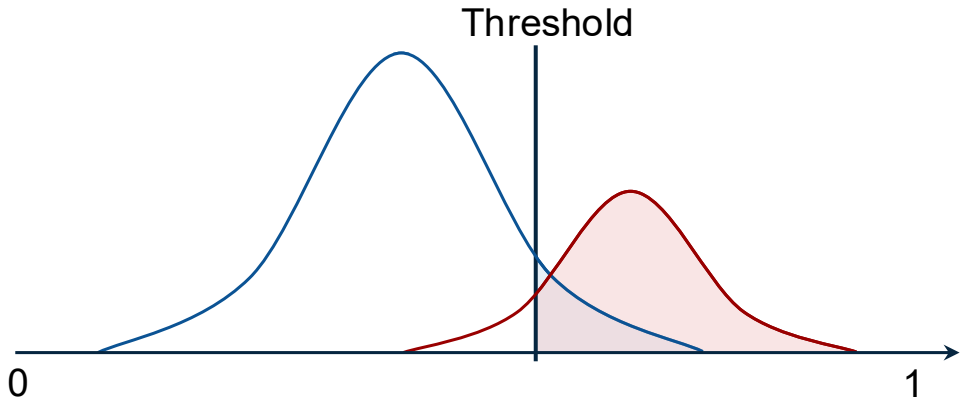


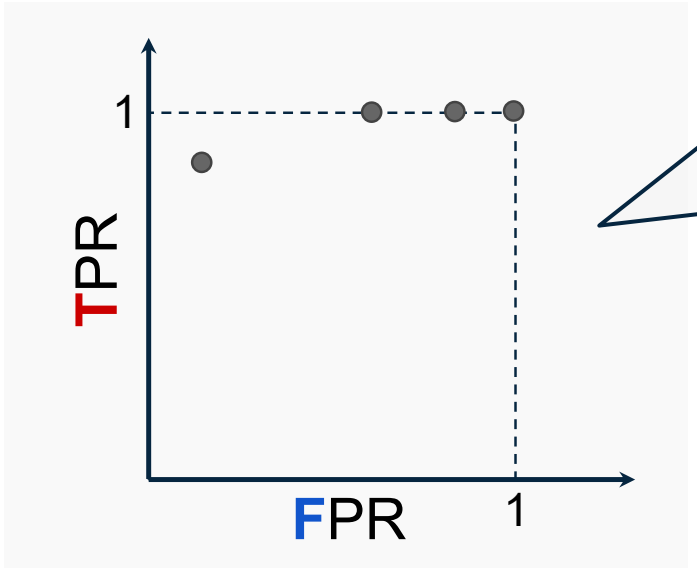


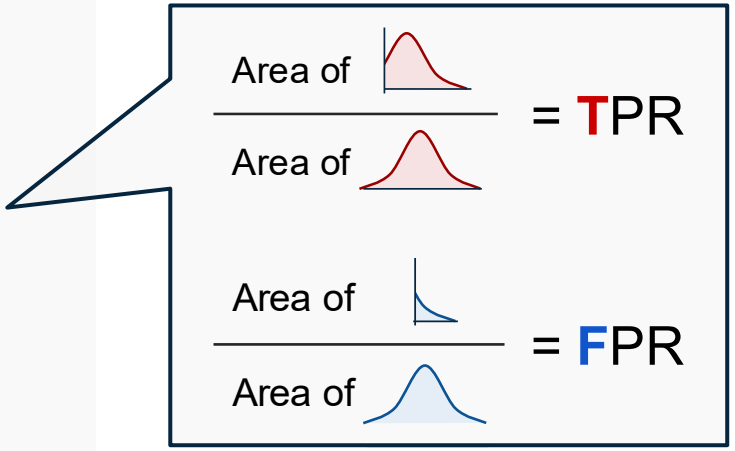
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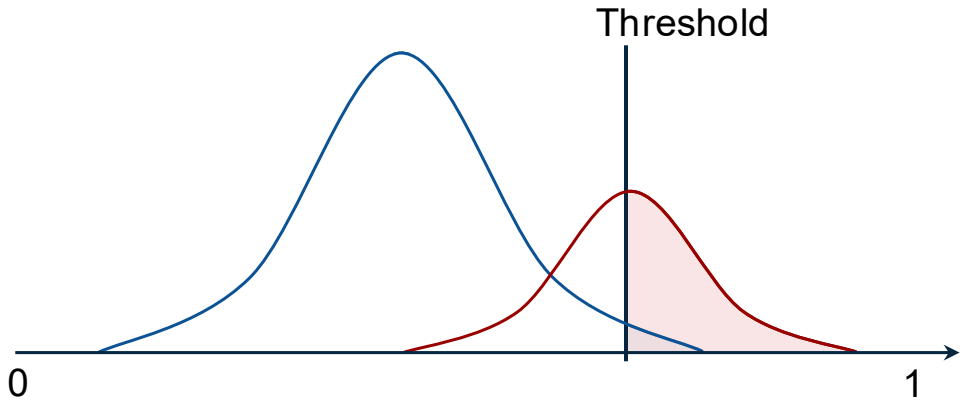


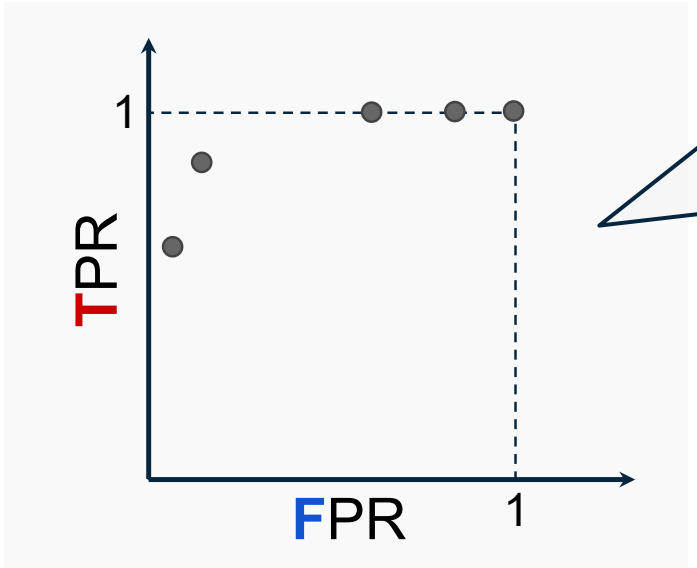


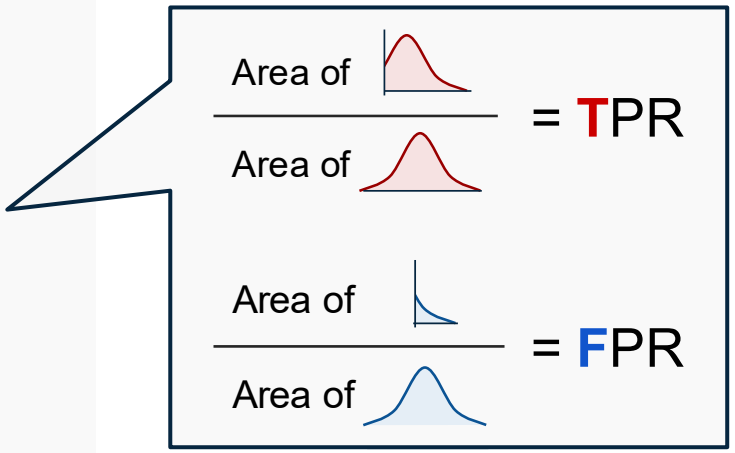
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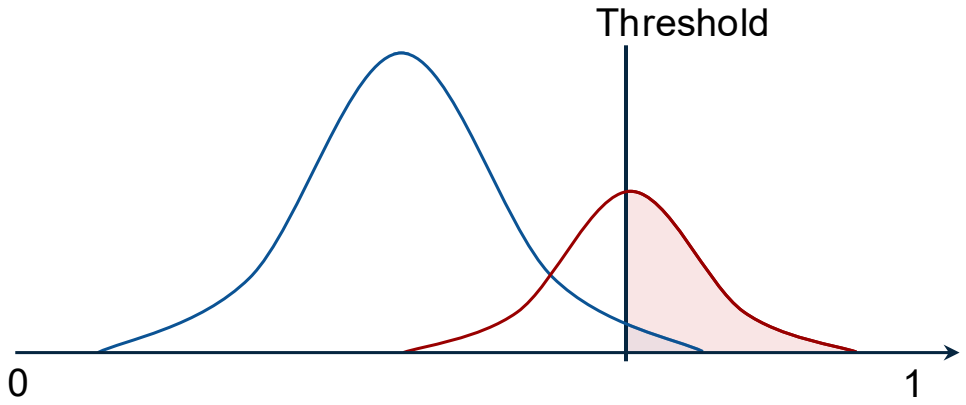


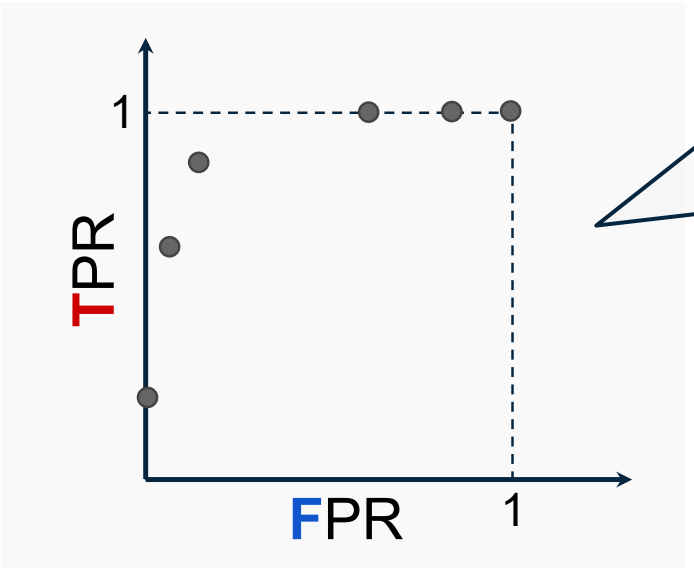


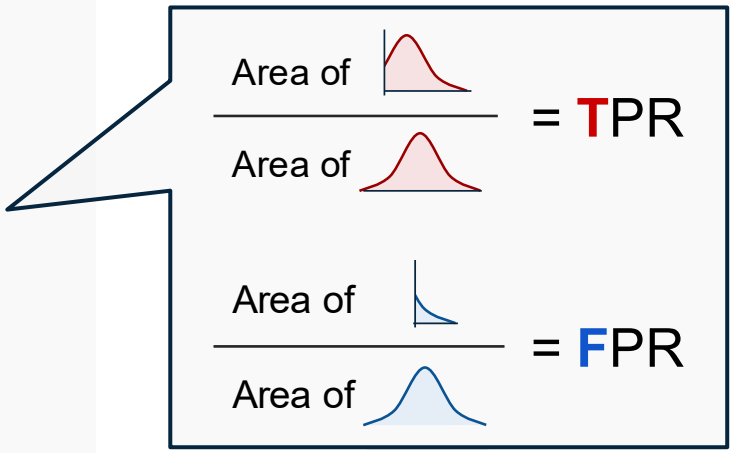
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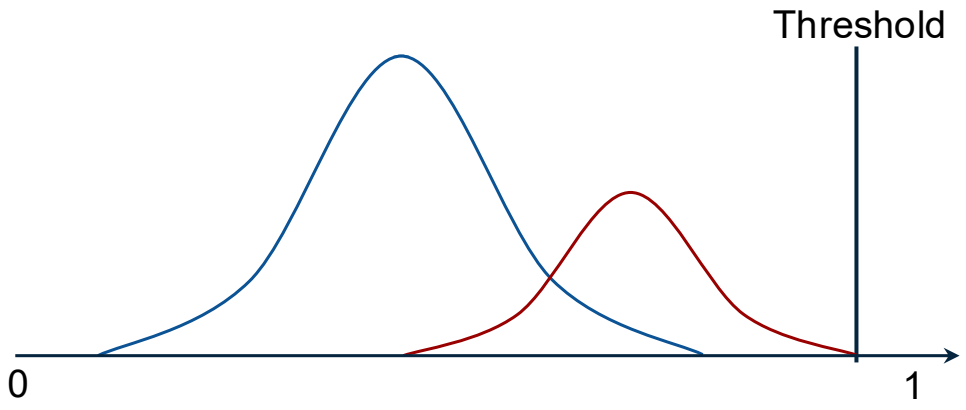


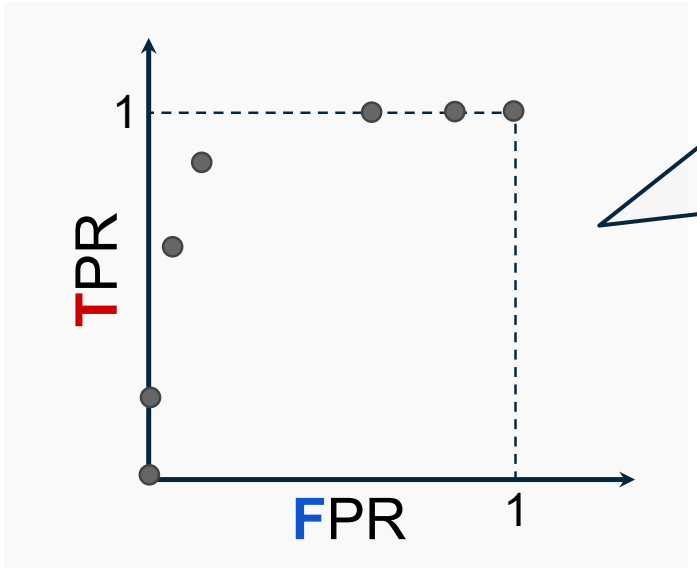


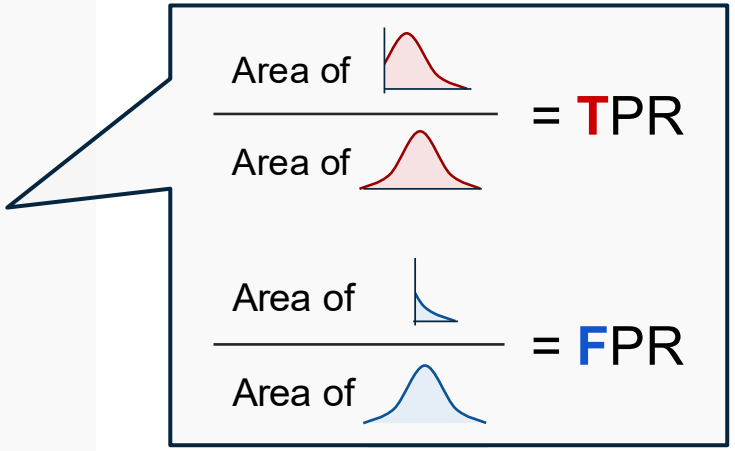
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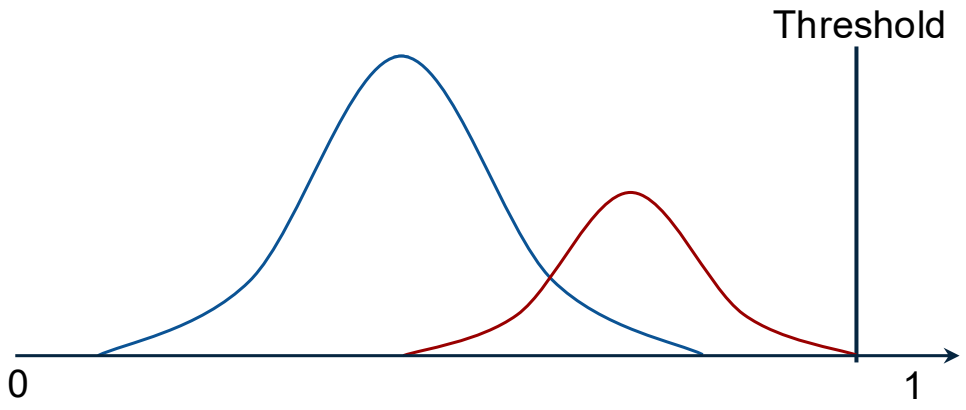


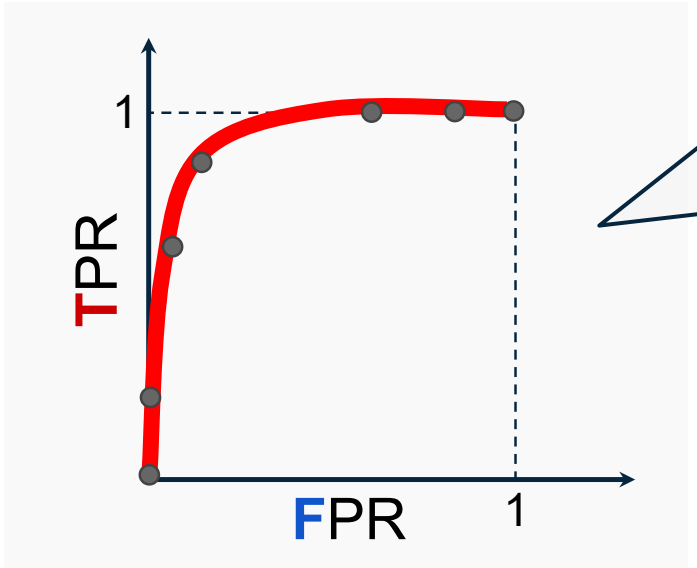


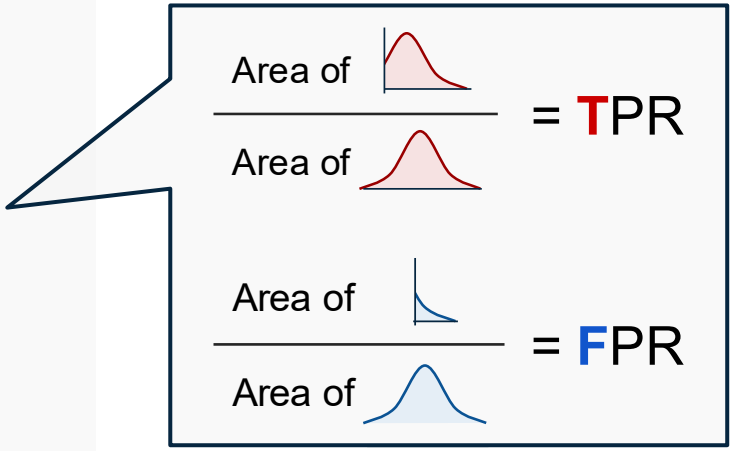
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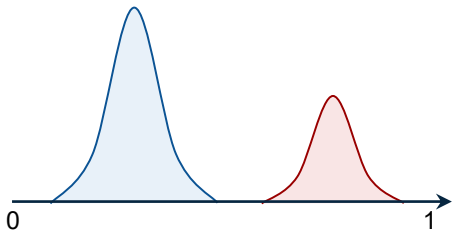


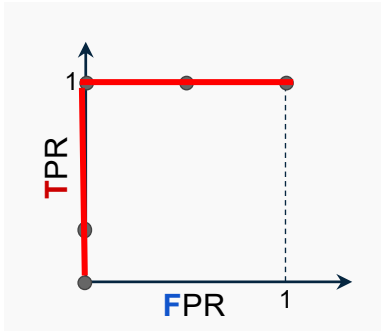
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Ideal Classifier & Guessing

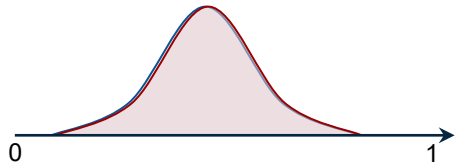
Ideal classifier

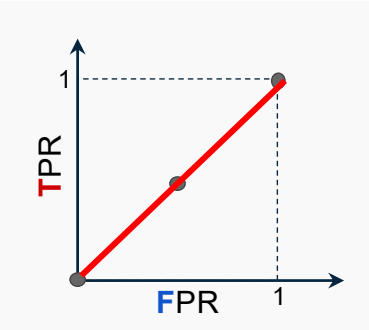




- With an ideal classifier, any positive sample has higher score than any negative sample
  → **the histograms do not overlap**
- ROC-AUC = 1

Guessing





- With half-and-half guessing,
  the histograms completely overlap
- ROC-AUC = 0.5

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Classification Model Evaluation Metrics: A Summary for Practical Use

Recall	• When **missed detections** are significantly higher risk than false positives (e.g., cancer screening, malware detection, disaster prediction, child abuse detection).
Precision	• When **false positives** are significantly higher risk than missed detections (e.g., spam, detection, decisions for irreversible procedures, purchasing speculative financial products).
F-Value(F1)	• When comparing models using a **comprehensive evaluation** and **Recall** and **Precision.**
ROC-AUC	• When comparing models with **imbalanced data** and evaluating overall performance across all thresholds. • When comparing across **different problem settings.**
Accuracy	• When you want a **quick, intuitive** accuracy assessment with balanced data. • When explaining complex evaluation metrics to a novice.

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How to implement

Probability of being positive

```
from sklearn.tree import
DecisionTreeClassifier

# set & train your model
clf = DecisionTreeClassifier()
clf.fit(X_train,y_train)
# prediction
y_pred = clf.predict_proba(X_test)

print(y_pred[:,1])
>>[0.910, 0.304, 0.603, 0.122,
0.730, ...]
```

ROC-AUC

```
from sklearn.svm import SVC
from sklearn.metrics import roc_auc_score

# set & train your model
model = SVC(probability=True)
model.fit(X_train, y_train)
# prediction
y_pred = model.predict_proba(X_test)[:,1]

roc_auc_score(y_test, y_pred)
# inputs: ground truth, prediction
>> 0.865544
```

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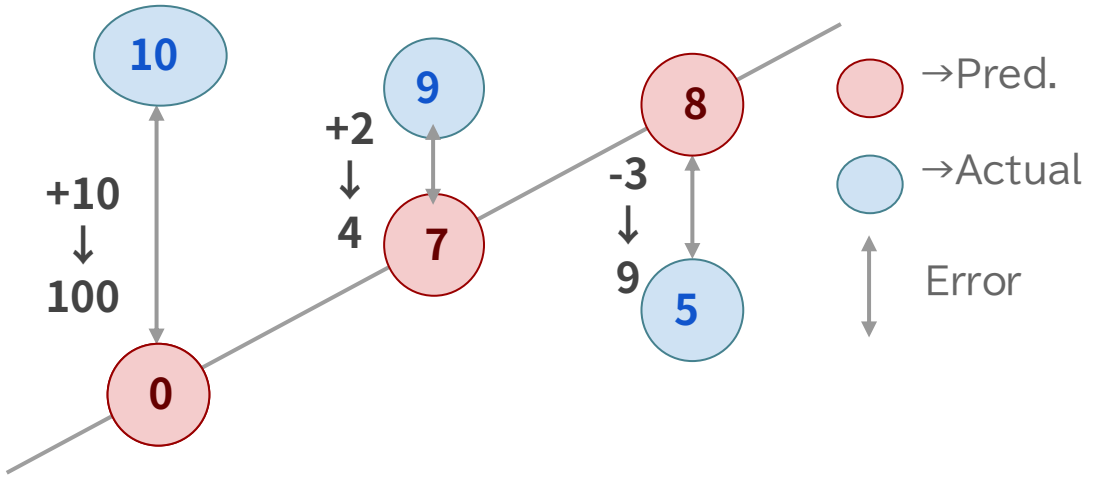
Evaluation for Regression Models

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Basic Idea of Evaluation Metrics for Regression

Error: key to evaluate regression

- Error = Actual − Prediction.
- Errors can be either positive or negative, so you can't simply sum them up (they will cancel out).
- The metrics are categorized based on how we handle these positive or negative differences to prevent them from canceling out.



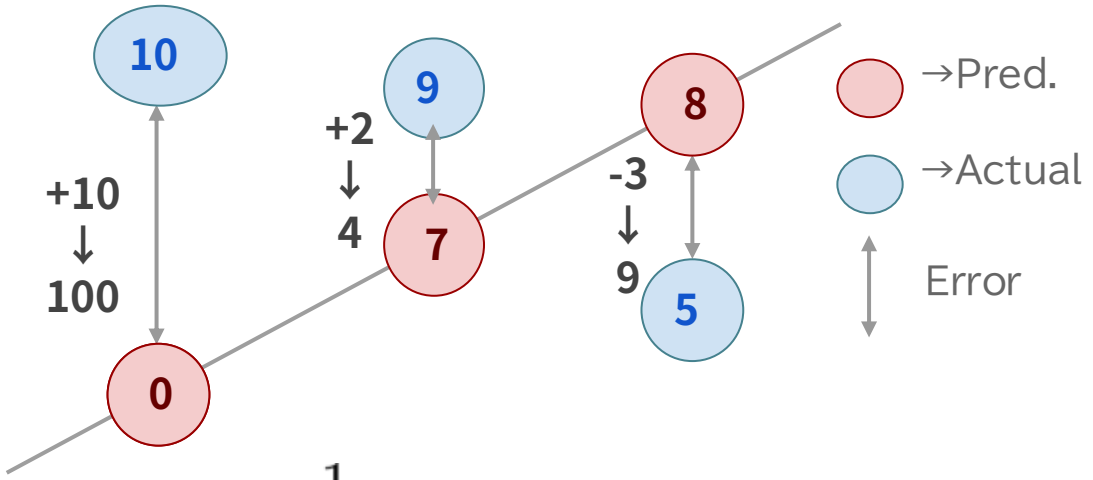
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Mean Squared Error (MSE)

MSE: Mean Squared Error

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|^2$$

- Objective function to minimize for regression models
- Penalize large losses more
- Prone to outliers



$$\mathrm{MSE} = \frac{1}{3} \cdot (100 + 4 + 9) \approx 37.7$$

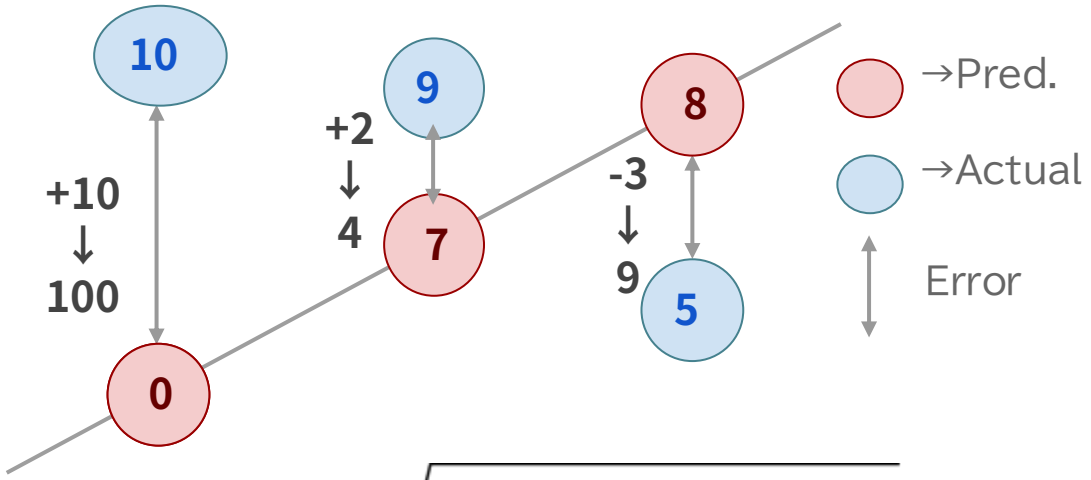
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Root Mean Squared Error (RMSE)

RMSE: Root Mean Squared Error

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|^2}$$

- Square root of MSE
- More intuitive than MSE since the unit matches



$$\mathrm{RMSE} = \sqrt{\frac{1}{3}\cdot(100+4+9)} \approx 6.1$$

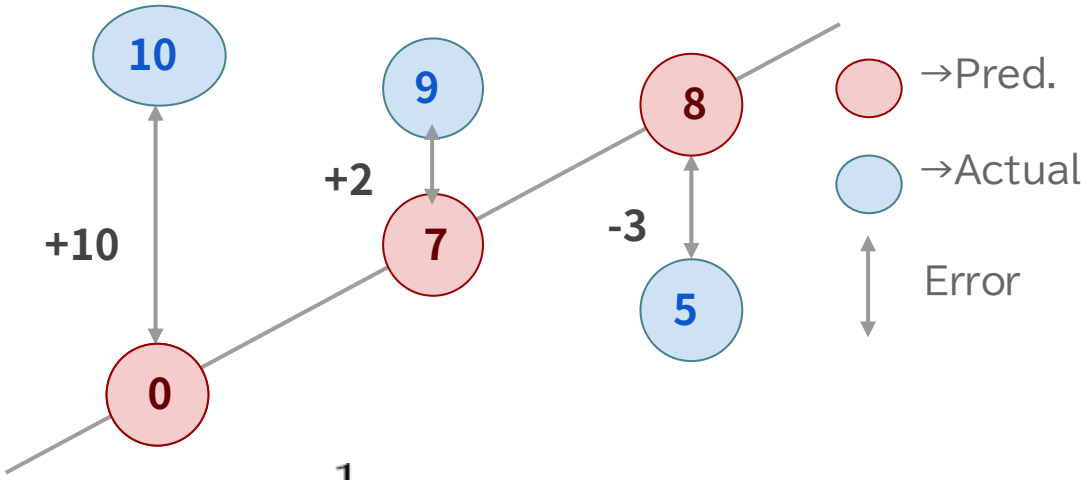
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Mean Absolute Error (MAE)

MAE: Mean Absolute Error

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

- Do not overrate the magnitude of errors
- Less susceptible to outliers than RMSE
- “average mistake”



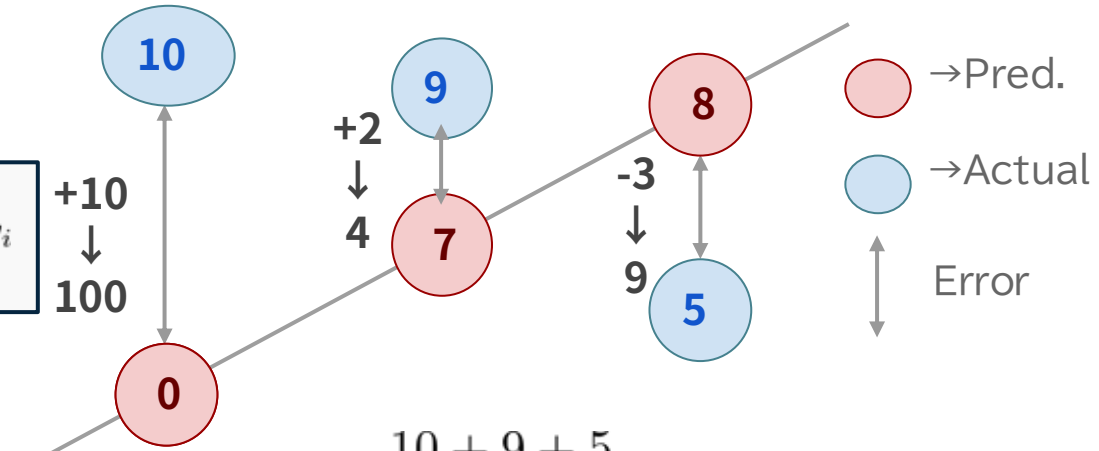
$$\mathrm{MAE} = \frac{1}{3}\cdot(|10| + |2| + |-3|) = 5$$

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Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{N} |y_i - \hat{y}_i|^2}{\sum_{i=1}^{N} |y_i - \bar{y}|^2}$$

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i$$



- Indicates how well the regression fits the data (≦ 1).
- As sensitive to outliers as MSE.
- It measures **whether the model reduces error compared with simply predicting the mean of the true values**.

$$\bar{y} = \frac{10 + 9 + 5}{3} = 8,$$

$$R^2 = 1 - \frac{100 + 4 + 9}{(10-8)^2 + (9-8)^2 + (5-8)^2}$$

$$\approx -7.07$$

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Regression Model Evaluation Metrics: A Summary for Practical Use

	Basic Properties	Usage	
MSE/RMSE	Evaluates models based on **avoiding large errors** rather than just overall average prediction quality.	For applications where losses increase significantly when prediction errors exceed a certain level (e.g., demand or investment forecasting).	The smaller, the better
MAE	Evaluates models based on **consistent average prediction quality** rather than the presence of large errors.	For applications where losses can be offset as long as predictions are accurate on average, even if some large errors are mixed in.	
R^2	**1.0 for perfect predictions, 0.0 for baseline** (guess using mean), and negative values for even worse results.	When you want to evaluate and compare model fit without being affected by the scale of the target variable.	The closer to 1, the better.

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How to implement

MSE, RMSE, MAE, and R^2

```
import numpy as np
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

y_pred = model.predict(X_test)

mse_score = mean_squared_error(y_test, y_pred)
rmse_score = np.sqrt(mse_score)
mae_score = mean_absolute_error(y_test, y_pred)
# Or root_mean_squared_error(y_test, y_pred) for sklearn>=1.4
r2 = r2_score(y_test, y_pred)
```

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Section Summary

- Understand & evaluate your model better by using various metrics.
- Depending on the type of your task, you can use different evaluation metrics
  - Classification: accuracy, recall, precision, F1, ROC-AUC, etc.
  - Regression: MSE, MAE, etc.
- Depending on the problem, different metrics may be adequate.

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